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List of Abbreviations

AIC	Akaike Information Criterion
(A)DF	(Augmented) Dickey-Fuller test
API	An arbitrary scale expressing the gravity or density of liquid petroleum products established by the American Petroleum Institute
CIS	Commonwealth of Independent States
CRDW	Cointegrating Regression Durbin-Watson test
CUSUM	Cumulative Sum of Recursive Residual
CUSUMSQ	Cumulative Sum of Squares of Recursive Residual
DTI	Department of Trade and Industry
d.o.f	Degree(s) of Freedom
DW	Durbin-Watson Statistic
ECM	Error Correction Model
EIA	Energy Information Administration
FIFO	First in, First out
FOB	Free on Board
HQC	Hannan Quinn Information Criterion
IC	Information Criterion
JB	Jarque-Bera
LIFO	Last in, First out
LS	Lest Squares
M-TAR model	Momentum-Threshold Autoregressive model
OFT	Office of Fair Trade
OLS	Ordinary Least Squares
p-value	Probability Value
RESET	Ramsey's REgression Specification Error Test
SBC	Schwarz Bayesian Information Criterion
SPS	Selective Price Support
TAR model	Threshold Autoregressive model
UK	United Kingdom

USA	United States of America

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Preface

In many markets the behaviour of the key features cannot be described as symmetric. Agents seem to respond in different ways to the incentives they experience, depending on the market situation. In this paper two examples of such behaviour are described.

The first (and the one author paid more attention) is concerned with well documented petrol marker in the UK. The second one shortly deals with the Polish pig market. The stress was put on the British market because many previous studies have dealt directly with it. Those studies were based only on the results of well-described ECM techniques, so the development of the latest techniques based on the TAR and M-TAR models gave author the opportunity to re-visit them and draw new conclusions.

Since so far no one analysed the asymmetries associated with the Polish pork market, the appendix is much shorter and only describes some general results.

I am indebted to my supervisor for his great supervision and advise provided during this dissertation. Many people offered very useful comments during the last few months, but I am especially indebted to Catarine Cosgrove and Chris Baker from the Institute of Petroleum, who provided me with details about the excise duties levied on petrol and additional data on North Sea crude oil composites. I am also indebted to Allan Hesse from OPAL for data on latest developments in the UK petroleum market.

During the creation of this work author was an employee of CASE-Doradcy and was heavily involved in conceptual and econometric projects connected with Polish petrol market. He was also involved in several econometric analyses for PKN Orlen®, but this work presents only his views and was not affected anyhow by his professional career. At the moment, the author is employed by Deloitte Touche Tohmatsu®. Also in this case the usual disclaimer applies.

Introduction

After sudden developments in the world oil markets many consumers argue that increases in crude oil prices are passed on to them quickly in the form of large and sudden increases in the price of petrol whilst decreases in the prices of crude oil lead to small decreases in the petrol prices.

It seems that consumers correctly expect prices of petrol to depend on prices of crude oil-main input used to produce it. What is more, they anticipate that increases and decreases of equal magnitude in the prices of input should lead to equal changes in output prices. It seems that they believe in the idea of perfect competition, in which firms use every opportunity to lower the prices and fear losing the market share by postponing price reductions.

The study of the asymmetric price transmission is quite matured, but every year brings new developments. Unfortunately, they are focused on the empirical part of the problem. Theoretical background is still neglected. Samuel Peltzman (2000:466) stated even that:

> ‘Output prices tend to respond faster to input increases than to decreases (....) It is found as frequently in producer goods as in consumer goods market (....) (it) suggest *a gap in the essential part of economic theory’* (emphasis added)

I do not even attempt to fill this gap, but I hope to establish whether petrol prices really responds faster to crude oil price increases than to decreases and find out which explanations might correctly identify what causes the asymmetry. Unfortunately this requires focusing mainly on the empirical work.

Another issue worth explaining is the market and country of interest. Petrol markets were the first analysed by researchers looking for asymmetries so it seems only natural to focus on those markets. It enables to compare the methodology used with other studies and forms a basis to show the supremacy of the new econometric techniques.

Most of the attempts to analyse petrol markets were done in the United Kingdom so also in this work the UK petrol market is under scrutiny. This gives the opportunity to compare the result obtained with previous studies and to draw more conclusions.

Author could not perform the similar study for Polish market because of several reasons. Firstly, public data is available only for less than 10 years, which is not enough to apply

sophisticated econometric techniques. Secondly, the methodology of data-collecting process has changed during that period. Thirdly, the illegal use of petroleum products is so common that in many cases it can seriously affect the results of any econometric study[1].

The cost-benefit analyses showed that it is better to analyse a mature market and to be able to compare results to other studies and to use more sophisticated techniques, than to deal with relatively short series for the Polish petrol market.

In this dissertation I deal with several issues. First I present up-to-date developments in the theory behind modelling asymmetric price transmission. I also report results of previous studies, both on UK and on other markets. Later I describe the empirical methods developed for assessing asymmetry in price transmissions. My aim is to provide the reader with knowledge necessary to understand the empirical analysis and point out why new techniques may be superior to those used before.

In the second chapter, I analyse the long-run relationship between crude oil prices and retail prices of 4 Star petrol. I start with the short description of the transmission chain, and issues connected with the Pricewatch campaign. I also analyse the data used, their advantages and disadvantages along with their statistical properties. The last part of this chapter describes the results of the application of time series techniques aimed at characterising the nature of the long-run relationship, its stability and problems that may be encountered at further stages of modelling.

Two last chapters deal directly with the issue of asymmetry and form the core of my work. My contribution to the topic is this empirical analysis of the transmission chain done with the use of two techniques:

- TAR and M-TAR models based on the concept of threshold cointegration;
- Enriched ECM which allows for short-run asymmetry.

Former technique allows me to get more insight into the nature of the asymmetry and analyse it in the different manner than previous researchers. Particularly, it offers a way to look at the way in which system adjusts to the long-run equilibrium depending on the characteristics of this disequilibrium and its effects on the market margins.

[1] The most important examples are illegal import of cheap petrol products or components from the CIS countries and illegal use of other kinds of oils (other than diesel) in the self-ignition engines. During his cooperation with PKN ORLEN author gathered significant knowledge of those issues and their impact on Polish market.

Latter technique is aimed at looking for the asymmetric patterns in the short run response, establishing timing of changes and their magnitudes. It also gives me the opportunity to search for the particular patterns in the asymmetric responses that may be consistent with some explanations of this phenomenon.

The last chapter focuses on those explanations. To the best of my knowledge, it is the first such attempt for the British petroleum market. I start with the informal discussion of all usual rationalisations. Later, I look whether empirical results seem to confirm them. In Peltzman's words (2000:488) I 'let the data speak'.

The appendix shows the application of the latest econometric techniques to the Polish pork market. It is important to mention that its main aim is to show that although TAR and M-TAR techniques were created to analyse the issues connected with the price transmission, they can be use to analyse other decisions made by the agents.

1 Theoretical Background

1.1 Review of the Literature

More than a dozen studies have specifically dealt with the issue of the asymmetric transmission of price changes in petroleum markets. Those studies differ in one or more of the following aspects:

- The country under scrutiny;
- The focus on national market, one region/town or several small markets;
- The time frequency of the data and period analysed;
- The focus on prices prevailing at different stages of the transmission chain (e.g. f.o.b. crude oil prices, prices of crude oil acquired by refineries, wholesale prices, retail prices, etc.);
- Dealing with upstream prices denominated in domestic currency or allowing for additional source of asymmetry in price transmission caused by movements in the exchange rate;
- The kind of petrol and crude oil analysed (regular, premium, leaded, unleaded, etc.);
- Taking into account the impact of taxes or using data net of all taxes;
- The model used in the empirical investigation;

Perhaps the first study to analyse the response of pump prices to crude oil cost changes was by Sumner (1990), Using Engle and Granger's two-step procedure he examined the relationship between net retail prices of 4 Star petrol, the producer price of crude oil acquired by refineries and labour costs (index of weekly earnings in manufacturing) over the sample February 1981 - December 1989.

The tests for cointegration were performed and existence of long-run relationship was established. For testing for the asymmetry he used the Wolfram-type segmentation of first differences of explanatory variables. The evidence of faster response of retail prices to increases as opposite to decreases in prices of crude oil was found. To check whether there was any offset through the ECM to the asymmetrical effects of cost changes the error correction term (lagged residuals from the level regression) was divided into Wolfram's manner. Results were not

significantly changed, which can be seen as evidence that over this period asymmetry was persistent.

The problem of a different response to price increases and decreases was also considered in Bacon's (1991) article. His work on the UK petrol market was focused on the transmission of spot crude oil prices (the ex-Rotterdam price was used as proxy for producer prices) to retail petrol prices. Biweekly data for the period 1982-1989 was used, and since the prices of crude oil were in U.S. dollars the impact of the exchange rate was also taken into consideration.

Evidence on asymmetry in transmission was found. Author concluded that increases in the crude oil prices are fully transmitted within 8 weeks while approximately one extra week is necessary for full transmission of the decreases in crude oil prices. What is more, changes in the exchange rate require two extra weeks relative to crude oil prices before being fully incorporated in retail prices.

Manning's (1991) work was inspired by the debate over the pricing policies used by major oil companies (so-called majors) for the retail price of petrol in the UK. He considered whether the retail prices of 4 Star petrol exhibit a long-run relationship with the level of Brent oil prices and excise duties using the techniques based on simple ECM. After ascertaining that the long-run relationship does exist[2] he considered whether the short-run adjustment of petrol prices to changes in the oil price is symmetrical or whether petrol retailers change petrol prices less rapidly and by lower amount in times of falling oil prices.

Monthly information from January 1973 to December 1989 was used to show that asymmetric pricing reactions by retailing companies was indeed present in the short-run although the difference between adjustment to increasing, as opposed to falling, oil prices was 'barely evident four months after the initial price change' (Manning 1991:1535).

He also concluded that approximately 7% of any deviation from the long run equilibrium was corrected or eliminated each month. The equation 3 on page 1539 with small and easy-to-correct mistake ($\Delta \ln p_{t-1}$ instead of $\ln p_{t-1}$) gives slightly greater speed of adjustment – 7.29% of disequilibrium is eliminated every month. His findings that disequilibria were persisted (only 60% was supposed to be eliminated after 12 months) were in contrast with other studies on the UK market.

[2] One reservation applies here: the t-statistic on error-correction term (lagged residual from level equation) has t-statistic equal to 2.45 only, which is obviously lower than critical values tabulated e.g. in Banerjee et al. (1993)), no other test for cointegration was reported.

For the US market, Karrenbrock (1991) employed 1983-1990 monthly data to study the empirical relationship between wholesale and retail petrol prices. Both prices used were reduced by the sum of the federal petrol tax and average of the 50 states' petrol tax.

The simple distributed lags model was used to find that the length of time in which a wholesale price increase is fully reflected in the retail petrol price is the same as that of a wholesale decrease for premium and unleaded regular petrol. The null hypothesis of symmetric transmission was not rejected. Author concluded that, contrary to commonly held belief that drivers do not benefit from wholesale petrol price decreases; these are passed along as fully and quickly as are wholesale price increases.

Another study concerned with asymmetric transmission in the USA was by Borenstein, Cameron and Gilbert (1997). For their analysis of price transmission between 1986 and 1992, they used weekly data on:

- Prices of West Texas Intermediate crude oil;
- Prices of generic petrol in New York and the Gulf Coast;
- Average price of branded petrol sold at terminals in 33 cities east of the Rocky Mountains;
- Average prices of unleaded regular self-service petrol net of all taxes in those cities.

They examined the properties of all series and found that all of them are first difference stationary. After testing for the homogeneity of the contemporaneous change in the upstream price in all of the transmissions they estimated, they proceeded with two-stage LS estimation.

Using asymmetric ECM, they confirmed that retail petrol prices respond more quickly to increases than to decreases in crude oil prices. The adjustment takes approximately 8 weeks in case of decrease in crude oil prices but only 4 weeks in case of increase in prices of crude oil.

By analysing price transmission at different points in the distribution chain they tried to distinguish between possible sources of asymmetry. They concluded that:

- Inventory adjustment effects may be to blame for asymmetry in the response of spot prices for generic petrol to changes in crude oil prices;
- Short run market power among retailers may be responsible for asymmetry in the response of net retail prices to wholesale price changes.

Details about those explanations will be provided in the last part of my dissertation.

Balke, Brown and Yücel (1998) extended the work of Borenstein et al. (1997) using different model specification. They used data from January 1987 through August 1996 on weekly prices of:

- West Texas Intermediate crude;
- Spot prices for unleaded petrol in the New York Harbour;
- Wholesale price of petrol and retail prices of self-service unleaded motor spirit with and without taxes.

After finding out that their series are stationary, they proceeded with analysis of Granger-causality and sources of variance. They established that upstream prices Granger-cause downstream prices at all stages of the distribution chain. The results of bivariate vector autoregressive models for each pair suggested that with one exception (relationship between spot price of crude oil and wholesale spot price of petrol) price shocks originate upstream and are transmitted downstream.

Their analysis was based on two techniques: distributed lag model in the levels of prices with asymmetric effects and ECM representation that allows for both long-run and short-run asymmetries. The results of the first model did not allow them to reject the null of symmetric response in all cases. But since the results were not systematic (Balke et al. (1998: 6) the second model, similar to that used in Borenstein et al. (1997), was estimated. The null of symmetry in transmission was rejected in nine of ten price pairs.

Reilly and Witt (1998) analysed the UK market and revisited the evidence of Bacon (1991) and Manning (1991) with monthly data for 1982-1995 stressing the importance of dollar-pound exchange rate and potential asymmetries associated with it, in addition to those of crude oil prices.

An unrestricted ECM, which allows for short-run asymmetries in adjustment to changes in the exchange rate and crude oil prices, was used and the evidence of asymmetries was found.

Authors concluded that the retail prices increase by more when the prices of crude oil increase, the same with the unfavourable changes in the exchange rate

Finally, some recent papers looked at different countries. Godby, Lintner, Stengos, Wandschneider (2000) used weekly data on self-service regular and premium patrol net of taxes and Edmonton par and Montreal Brent crude oil cost for the period January 1990 to December 1996 for 13 Canadian cities. They applied Hansen's bootstrap procedure to test the null

hypothesis of a linear formulation against an asymmetric alternative. The test was based on bootstrap critical values of a Wald type heteroskedasticity-consistent tests. Only weak signs of asymmetry were found and authors stated that asymmetry was not present (Godby et al. (2000: 364).

Authors conclude that it may be due to the differences in the market structure between Canada and other countries (fall in the petrol consumption per vehicle, large number of small stations), to the frequency of the data used and to the fact that previous studies used aggregate data on prices from distinct regions.

The only study known to the author that examined the issue of interest in more than one country was by Galeotti, Lanza and Manera (2002). They used monthly data (from 1985 to 2000) on prices at two stages of transmission (first refinery stage and second distribution stage) for five European countries (Germany, France, UK, Italy and Spain) to test for asymmetric transmission. They also allowed for asymmetries caused by the exchange rate.

The enriched, asymmetric ECM was used to distinguish between asymmetries that arise from short-run deviations in input prices and from the speed at which the petrol price reverts to its long-run level. Statistical inference was based on bootstrapped F-tests of asymmetries in order to overcome the low-power problem of conventional testing procedures. The results indicated widespread differences among European Countries in both adjustment speeds and short run responses to crude oil prices rises and falls.

Transmission from crude oil prices to retail petrol prices in the UK was found to be asymmetric. Detailed analysis of the second stage of transmission (from spot wholesale price to retail prices) indicated that also on this stage that asymmetries in the short-run responses and adjustment speed were present

1.2 Theory behind Testing for Asymmetric Price Transmission

1.2.1 Asymmetric ECM

The first model aimed at analysing the issue of asymmetric price transmission was developed by Houck in 1977. Many authors have used approaches for analysing price transmission, which differ in some technical regards but nonetheless can subsumed under the general title "the Houck approach'. The typical model from this class can be described as follows:

Consider transmission between two stages in the chain. For simplicity let's assume that upstream prices (e.g. crude oil prices) are expressed in the same currency as downstream prices (e.g. petrol) and the upstream price is the only factor affecting the downstream price.

Using the difference operators - Δs (Δ^+ stands for the product of the first difference and a dummy variable which takes the value of unity when appropriate first difference is positive, opposite for Δ[3]) the simplest "Houck-type' model can be written as:

$$\Delta \ln(\mathrm{p})_t = \alpha + \beta^+ * \Delta^+ \ln(c)_t + \beta^- * \Delta^- \ln(c)_t + \varepsilon_t \qquad (1)$$

By assumption, causation comes from the upstream and shocks to downstream prices are not transmitted upstream.

Testing for the asymmetry is constrained to normal tests (e.g. F-tests) of the null hypothesis of equal coefficients (H_0:β^+=β^-). If null is rejected the conclusion is that retail prices respond in different ways when upstream prices increase and decrease.

In the 1970s and early 1980s many authors used modelling techniques based on (1) to test for the asymmetry in the price transmission (see Von Crammon-Taubadel and Meyer (2001) for excellent review). However, along with the developments in the theory of time-series modelling, testing asymmetry by using extensions of (1) was more and more criticised.

As pointed out by von Cramon-Taubadel and Loy (1999) and von Cramon-Taubadel (1998), when modelling transmission by using (1) researchers did not utilise the long-run information. This technique is based only on the extended VAR in first differences and completely ignores the information reflected in levels of variables. Such models do not allow for

[3] Such notation is used in the rest of my dissertation

thorough testing for cointegration; do not give the estimates of the speed of adjustment to shocks and long-run coefficients of pass-through.

Since the early 1990s the tendency to use more sophisticated methods for analysis of prices transmission was clearly visible. The main problem was to reconcile straightforward testing for asymmetry possible in (1) with taking account for the existence of the long-run relationship between prices.

Bacon (1991) was one of the first to reconcile those two objectives. His approach can be explained as follows. Suppose again that the crude cost is the single factor in petrol prices and a stable long-run relationship in the following form (for simplicity the time trend is omitted) is present over the sample period:

$$\ln(p)_t^{\sim} = \alpha + \beta_1 * \ln(c)_t + \beta_2 * \ln(ex)_t + \varepsilon_t \qquad (2)$$

Where $ln(p)^{\sim}$ stands for the long-run equilibrium price, α is supposed to capture other costs, margins, effects of differences in the heat content, etc.. The β_1 coefficient stands for the elasticity of petrol price with respect to prices of crude oil expressed in USD, β_2 represents elasticity with respect to the exchange rate and ε represents the error term. If the exchange rate affects the price of petrol only through the price of crude oil (i.e. $\beta_1=\beta_2$) one can use crude oil prices expressed in national currency, so the equation (2) becomes:

$$\ln(p)_t^{\sim} = \alpha + \beta * (\ln(c)_t + \ln(ex)_t) + \varepsilon_t \qquad (3)$$

In such setting, according to Bacon (1991), it is possible to explain the response of petrol prices to input cost shocks in a simple non-linear partial adjustment model.

In such specification the current price - $ln(p)_t$ is set at last period's price with an adjustment between last period's price and the target level $ln(p)^{\sim}_{t-1}$ according to the following relationship:

$$\ln(p)_t = \ln(p)_{t-1} + \pi * (\ln(p)_{t-1}^{\sim} - \ln(p)_{t-1}) + \varepsilon_t \qquad (4)$$

In such setting π represents the speed of adjustment, under assumption that the downstream price adjusts in the same manner to positive and negative changes in the upstream price. He suggested that the following quadratic partial adjustment mechanism based on (4) should be used to test for asymmetries in price transmission

$$\ln(p)_t = \ln(p)_{t-1} + \pi_0 * (\ln(p)_{t-1}^{\sim} - \ln(p)_{t-1}) + \pi_1 * (\ln(p)_{t-1}^{\sim} - \ln(p)_{t-1})^2 + \varepsilon_t \tag{5}$$

In (5) the null hypothesis of π_1=0 reflects a symmetric pricing response. If both π_0 and π_1 are statistically greater than zero, the response to a crude cost increase will be more rapid early on than to a crude cost decrease. If π_0<0 and π_1>0, the response to a crude cost decrease is more rapid.

His approach has its drawbacks. The partial adjustment model assumes equal proportional adjustment towards new equilibrium in all periods after the shock. What is more, in the quadratic partial adjustment mechanism it is assumed that the asymmetry becomes proportionately larger with an increase in the difference between current and long run equilibrium prices (see Borenstein et al. (1997) for excellent discussion).

This constraint imposed on the nature of asymmetry is so serious that Bacon's (1991) technique will not be used in my dissertation.

The alternative way to analyse asymmetric price transmission is based on Error Correction Representation. If first-difference stationary time series are cointegrated (there is a long run relationship between them) then the ECM does exist and, in the most general form without the deterministic trend, can be depicted as:

$$\begin{aligned} \Delta \ln(p)_t &= \alpha_0 + \sum_{i=0}^{M_1} \beta_{1,i} * \Delta \ln(c)_{t-i} + \sum_{i=0}^{M_2} \beta_{2,i} * \Delta \ln(ex)_{t-i} + \\ &\sum_{i=1}^{M_3} \beta_{3,i} * \Delta \ln(p)_{t-i} + \pi_1 * \hat{u}_{t-1} + \varepsilon_t; \\ &\hat{u}_{t-1} = \ln(p)_{t-1} - \hat{\alpha} - \hat{\beta}_1 * \ln(c)_{t-1} - \hat{\beta}_2 * \ln(ex)_{t-1} \end{aligned} \tag{6}$$

or:

$$\begin{aligned} \Delta \ln(p)_t &= \alpha_0 + \sum_{i=0}^{M_1} \beta_{1,i} * \Delta \ln(c)_{t-i} + \sum_{i=0}^{M_2} \beta_{2,i} * \Delta \ln(ex)_{t-i} + \sum_{i=1}^{M_3} \beta_{3,i} * \Delta \ln(p)_{t-i} + \\ &\pi_1 * \ln(p)_{t-1} + \pi_2 * \ln(c)_{t-1} + \pi_3 * \ln(ex)_{t-1} + \varepsilon_t \end{aligned} \tag{7}$$

Where *M*s stand for the number of lags over which the shock is felt and ε represents the error term, which should be white noise. It is explicitly assumed that both explanatory variables are exogenous. In (6) and (7) the coefficients π_1 represents the estimate of the speed of adjustment of the downstream prices to shocks in the upstream prices. It is important to add that

in equations (6) and (7) it is implicitly assumed that the speed of adjustment is the same in case of positive and negative changes in the upstream prices.

It is possible to use ECM in testing for asymmetry in two ways, choice of the technique depends on whether researcher is interested in the speed of adjustment to the long-run equilibrium or short-run response to upstream price changes.

First way used by e.g. Borenstein et al. (1997) and Reilly and Witt (1998) is aimed at looking for the patterns in the short-run response like those depicted in Figure 1.

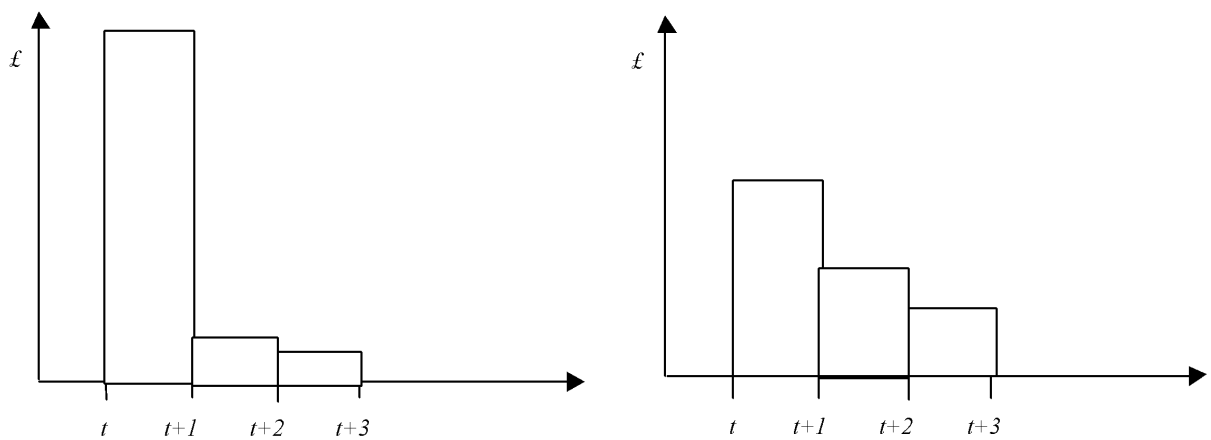

Figure 1 Example of asymmetry in the short-run responses.

Absolute values of changes in the petrol prices after increase and decrease of equal magnitude in crude oil prices at time t. Left panel shows responses to increases in crude oil prices.

It is based on separation of changes in the explanatory variables into positive and negative parts so that (7) becomes:

$$\Delta \ln(p)_t = \alpha_0 + \alpha_1 * trend + \sum_{i=0}^{M_1} \beta_{1,i}^{+} * \Delta^{+} \ln(c)_{t-i} + \sum_{i=0}^{M_2} \beta_{1,i} * \Delta \ln(c)_{t-i} + \sum_{i=0}^{M_3} \beta_{2,i}^{+} * \Delta^{+} \ln(ex)_{t-i} + \sum_{i=0}^{M_4} \beta_{2,i} * \Delta \ln(ex)_{t-i} + \sum_{i=1}^{M_5} \Delta \ln(p)_{t-i} + \pi_1 * \ln(p)_{t-1} + \pi_2 * \ln(c)_{t-1} + \pi_3 * \ln(ex)_{t-1} + \varepsilon_t \quad (8)$$

Notation used is the same as before. To ensure that the error term is indeed white noise, Borenstein et al (1997) suggested including lagged changes in the past petrol price. Reilly and Witt (1998) suggested including the deterministic trend so as to capture the effects of other costs (Reilly and Witt (1998: 299).

The number of lags in the estimation may be determined by Hendry's 'general-to-specific procedure' as in Reilly and Witt (1998) or specific-to-general procedure as in Borenstein et al.

(1997). Hendry's approach is preferred nowadays, since it allows for determining the overall size of the testing undertaken.

The coefficients on Δ^+s provide an estimate of the differential in the effect on net retail petrol prices of increases and decreases in crude oil prices or in the exchange rate. Those coefficients and their standards errors provide the basis for a statistical test of the symmetry hypothesis. The asymmetry is present when the coefficients on Δ^+s are statistically different from zero in the final model. The significance level used for testing the null that those coefficients are equal to zero is usually 5%.

The sign of those coefficients determines the direction of asymmetry in the response of net retail petrol prices to changes in the costs of producing petrol. For example, if the coefficient on contemporaneous positive first difference in crude oil price ($\beta^+_{1,0}$) is positive and statistically different from zero one can say that immediate response of the net retail petrol prices is greater for increases in the crude oil prices than for decreases. Analogous reasoning can be applied to the interpretation of β_2 and β^+_2 - exchange rate coefficients.

Technique similar to that described by equation (8) was used in Manning (1991). For testing the hypothesis that the reaction of petrol prices to crude oil price changes is symmetrical he amended (8) by introducing intercept dummy, which equals unity if the crude oil price increased in the particular month. However, his method involved excluding the constant (α_0) from (8) in the 'general-to-specific' procedure and this usually is not recommended in the literature.

Alternative specification of the ECM, given by equation (6) may be useful when it comes to estimation of the speed of adjustment to the long-run equilibrium after shocks to upstream prices.

Testing of such hypothesis is impossible in the framework discussed so far, since system was assumed to adjust to positive and negative shocks in the same manner.

One way to overcome this drawback was proposed by Granger and Lee (1989). They propose a modification to equation (6) that involves a Wolfram-type (1971) segmentation of the error-correction term (i.e. lagged residuals) into its positive and negative components:

$$\begin{aligned}
\Delta \ln(p)_t &= \alpha_0 + \alpha_1 * trend + \sum_{i=0}^{M_1} \beta_{1,i} * \Delta \ln(c)_{t-i} + \sum_{i=0}^{M_2} \beta_{2,1} * \Delta \ln(ex)_{t-1} + \\
&\sum_{i=1}^{M_3} \beta_{3,1} * \Delta \ln(p)_{t-1} + \pi_1^+ * \hat{u}_{t-1}^+ + \pi_1^- * \hat{u}_{t-1}^- + \varepsilon_t; \\
&\hat{u}_{t-1} = \ln(p)_{t-1} - a_0 - \alpha_1 * trend - \beta_1 * \ln(c)_{t-1} - \beta_2 * \ln(ex)_{t-1}; \\
&\hat{u}_{t-1}^+ = \max(0, \hat{u}_{t-1}); \\
&\hat{u}_{t-1}^- = \min(0, \hat{u}_{t-1});
\end{aligned} \tag{9}$$

It is usually recommended to estimate (9) after testing for cointegration in "traditional' way without splitting residuals into positive and negative parts. For testing whether the transmission is symmetric and downstream prices adjust equally quickly to increases and decreases in upstream prices von Cramon-Taubadel and Meyer (2001) proposed simple F-test of the null hypothesis of equal coefficients on positive and negative residuals (H_0:$\pi^+ = \pi^-$). If the null is rejected and $|\pi^+| < |\pi^-|$ the downstream prices adjust more rapidly to the increases in the upstream prices than to decreases so that the long-run equilibrium price is reached much faster.

Cook et al. (1998) argued that coefficients on these partitioned terms must not only be significantly different from each other but also must both be significantly different from zero. More recently, Cook (1999) applied this rule to the data originally used by Granger and Lee (1989) and showed that the additional requirement is very harsh (see Cook (1999) for details). In many studies about the price transmission based on Wolfram-type segmentation, this problem was visible (e.g. Abdulai (2002), Galeotti et al. (2002)) but neglected.

1.2.2 Asymmetric Time-Series Models

While the asymmetric ECM approach, as described by (8) and (9), may be superior to earlier approaches when the price series in question are cointegrated, the Monte Carlo experiments reported by Pippenger and Goering (1993), Balke and Fomby (1997) and Enders and Granger (1998) demonstrated that tests for unit roots and cointegration have some serious drawbacks in the presence of asymmetric adjustment.

They suggest that when adjustment towards new long run equilibrium is asymmetric and the degree of asymmetry is significant, the threshold-autoregressive (TAR) and momentum-TAR models should be used (see Enders and Siklos (1999: 23) for power comparisons.

Those techniques are especially useful when researchers are interested in analysing the process of adjustment of downstream prices after the shocks to upstream prices like those depicted in Figure 2

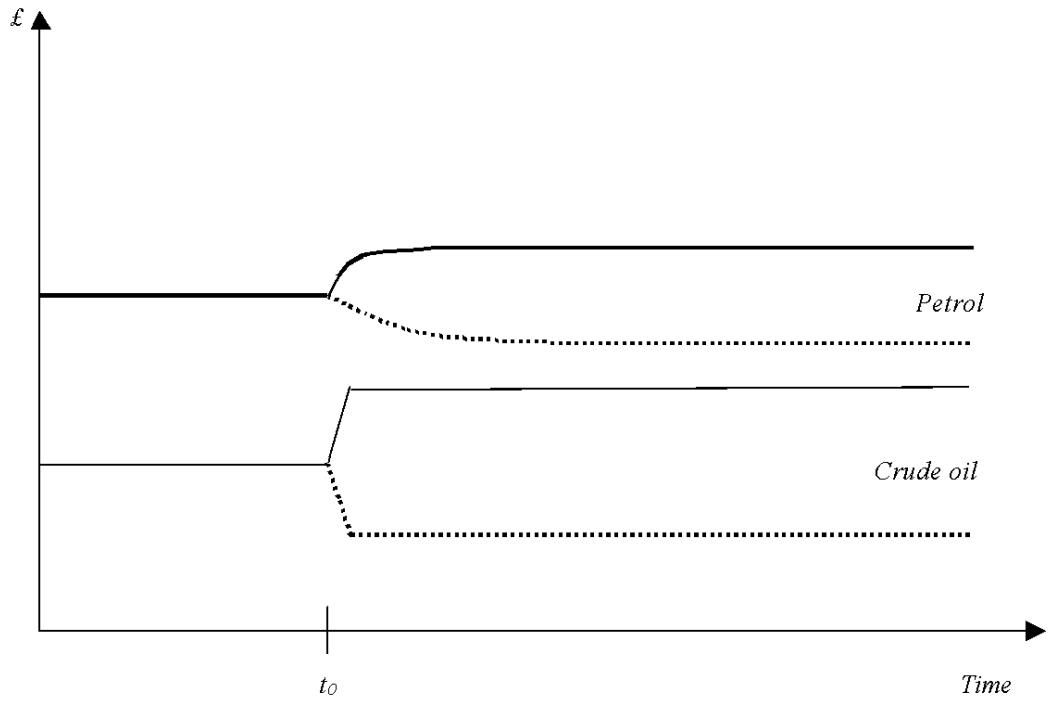

Figure 2 Example of asymmetry in the long-run speed of adjustment.

Process of adjustment of petrol prices after the increase and decrease in crude oil prices at time t_0. Dashed line represents adjustment after a decrease in crude oil prices.

They are aimed at capturing the difference between the speeds of adjustment of downstream prices (petrol prices) to negative (caused by increases in crude oil prices) and positive (caused by increases in crude oil prices) disequilibria.

Mentioned alternatives to conventional techniques can be seen as an extension to the standard Engle-Granger (EG) testing strategy. The EG method assumes linearity and symmetric adjustment. In the simplest case, this two-step methodology entails:

1) Using OLS to estimate the long-run relationship as:

$$\ln(p)_t = \alpha + \beta * (\ln(c)_t + \ln(ex)_t) + u_t \qquad (10)$$

Where all variables are the individual I(1) components, α and βs are the estimated parameters and u_t is the disturbance term that can be correlated.

2) OLS estimation of the π in the regression equation:

$$\Delta\hat{u}_t = \pi * \hat{u}_{t-1} + \varepsilon_t \qquad (11)$$

Where ε_t is a white noise disturbance and the residuals from (10) are used to estimate (11). Rejecting the null hypothesis of no cointegration (i.e. accepting the alternative hypothesis: $\pi \in]-2;0[$, implies that the residuals in (10) are stationary with mean zero. As such (10) represents long-run equilibrium - attractor such that its pull is strictly proportional to the absolute value of u_t.

It is important to add that Balke and Fomby (1997) and Enders and Siklos (1999) argued that for a small degree of asymmetry, which is typical in many economic applications, the EG method described by (11) work reasonably well (Balke and Fomby (1997: 627). Therefore it can be used to detect cointegration when asymmetric adjustment is present. However, to get more insight into the characteristics of the process governing the adjustment towards long-run equilibrium alternative techniques (TAR and M-TAR) should be used.

Enders and Siklos (1999) and Enders and Granger (1996) proposed that the following equation (TAR) should be estimated:

$$\Delta\hat{u}_t = \pi_1 * I(\hat{u}_{t-1}) * \hat{u}_{t-1} + \pi_2 * (1 - I(\hat{u}_{t-1})) * \hat{u}_{t-1} + \varepsilon_t \qquad (12)$$

Where $I(u\hat{}_{t-1})$ is the indicator function (indicator in short) such that:

$$I(\hat{u}_{t-1}) = \begin{cases} 1 \Leftrightarrow \hat{u}_{t-1} > \tau \\ 0 \Leftrightarrow \hat{u}_{t-1} < \tau \end{cases} \tag{13}$$

The value of the threshold is represented by τ. Setting the threshold at the level different from zero is the only difference to equation (11).

The necessary and sufficient condition for the stationarity of $\{u_t\}$ in such case is: $(\pi_1<0) \wedge (\pi_2<0) \wedge [(1+\pi_1)*(1+\pi_2)<1]$ for given values of τ von Cramon-Taubadel and Meyer (2001:6). If this condition is met, u_t=0 can be considered the long-run equilibrium value of the system. The adjustment of the system in such case is $\pi_1 * u_{t-1}$ when $u_{t-1} > \tau$ (e.g. after sudden and large decrease in the upstream prices) and $\pi_2 * u_{t-1}$ in the opposite case.

Enders and Siklos (1999) tabulated critical values of an F-test (so-called ϕ* test, or just ϕ when threshold equals zero) that can be used to test join hypothesis that $\pi_1 = \pi_2 = 0$[4]. If this hypothesis is rejected we can conclude that the cointegration between variables of interest is present.

According to Balke and Fomby (1997:4), this test can be seen as additional tool for testing for presence of the long-run relationship. Enders and Granger (1996) also tabulated critical values for so-called t-max* test (or just t-max when threshold is assumed to be equal zero). It is based on maximum t-statistic of estimate of π_1 or π_2. If the less negative t-statistic exceeds the critical values one can conclude that long run relationship between variables exists.

If the existence of long-run relationship is established, following Tong (1990), who showed that the OLS estimates of π_1 and π_2 have an asymptotic multivariate normal distribution, it is possible to use standard F-test of the null hypothesis that $\pi_1 = \pi_2$ in order to test whether adjustment is symmetric. When null is rejected and an alternative hypothesis that $|\pi_1| < |\pi_2|$ is accepted, one can conclude that downstream prices adjust to the new long-run equilibrium faster when crude oil prices increase than when crude oil prices decrease.

In general, the true value of τ is unknown and needs to be estimated. In a number of economic applications it seems natural to set $\tau=0$ so that the attractor is just the cointegrating vector. When possibility of threshold other than zero is real, Enders and Siklos (1999) suggest utilising so-called "Chan's approach'.

4

Chan (1993) showed that, under some regularity conditions, the least squares (LS) estimator of a stationary ergodic threshold autoregressive model is strongly consistent. After deriving the limiting distribution of the OLS estimator of the threshold parameter he showed that it is N consistent (Chan (1993: 520) so searching over the potential threshold values so as to minimise the sum of squared errors from the fitted model yields a super-consistent estimate of the threshold.

Usual procedure, employed e.g. in Abdulai (2002), is based on sorting estimated residual series from (10) in ascending order, discarding the largest and the smallest 15% and considering each of the remaining 70% as possible thresholds. For each of the possible threshold (13) is recomputed and (12) or (14) is estimated. The threshold yielding the lowest residual sum of squares in (12) or (14) is deemed to be the appropriate estimate. Harris and Silverstone (1999) suggested using the percentiles of the residuals, which is less computation-burdensome.

The basic model described above (with or without threshold set at zero) can be modified in many ways. The one usually advised involves augmenting (12) with lagged changes in the $\{u_t\}$ sequence such that it becomes the k-th order process:

$$\Delta\hat{u}_t = \pi_1 * I(\hat{u}_{t-1}) * \hat{u}_{t-1} + \pi_2 * (1 - I(\hat{u}_{t-1})) * \hat{u}_{t-1} + \sum_{i=1}^{k-1} \gamma_i * \Delta\hat{u}_{t-i} + \varepsilon_t \tag{14}$$

In working with specifications such as (14), various model selection criteria (such as AIC or SBC) and diagnostic checks of the residuals (such as the autocorrelogram of the residuals and Ljung-Box tests) can be used to determine the appropriate lag length (again see Tong (1990) for excellent discussion). All tests described above can be used, but the critical values appropriate for ϕ test are slightly different (Enders and Siklos (1999))

In (12) and (14), the indicator depends on the level of lagged residuals from the long-run equation. Caner and Hansen (1998) and Enders and Granger (1998) suggested an alternative such that threshold depends on the previous period's *change* in residuals so (13) becomes:

$$I(\hat{u}_{t-1}) = \begin{cases} 1 \Leftrightarrow \Delta\hat{u}_{t-1} > \tau \\ 0 \Leftrightarrow \Delta\hat{u}_{t-1} < \tau \end{cases} \tag{15}$$

Such models are called momentum-threshold autoregressive (M-TAR) models in that the $\{u_t\}$ series exhibits more "momentum' in one direction than the other. Of course setting τ=0 in many cases is natural and is recommended. Including lagged changes in the $\{u_t\}$ sequence, just as in (14) is allowed. What is more, utilizing Chan's procedure for finding estimate of the threshold is possible.

Testing for cointegration involves the F-test (this time named ϕ*(M) test or ϕ (M) if threshold equals zero) of the join hypothesis that $\pi_1 = \pi_2 = 0$ and comparing it with the critical values tabulated again in Enders and Siklos (1999). Also t-max tests are allowed (they were called t*-max(M)). Again in is important to add that for reasonable degree of asymmetry conventional tests work reasonably well.

The figure below can serve as an example of difference between 'normal' and consistent versions of TAR and M-TAR model. On the vertical axis is the change in the disequilibrium - the adjustment in the period right after the shock, and on the horizontal axis the magnitude of the shock. It is clear that in consistent version of the TAR model the adjustment to large, positive shocks is larger than to small shocks.

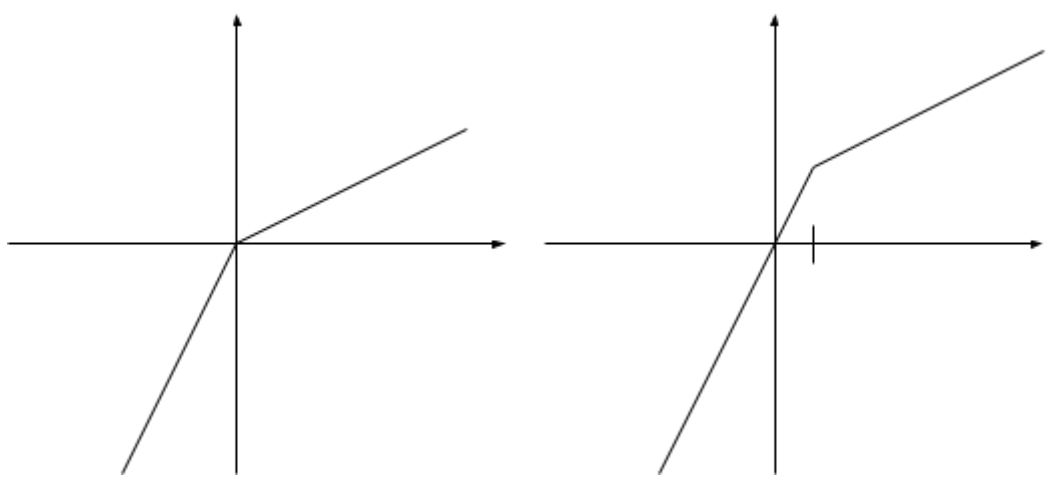

Figure SEQ Figure * ARABIC 3 TAR models with threshold set at 0 (left panel) and with positive threshold (right panel).

Before proceeding with the properties of the model it is important to add that they are not fully developed. For example, only one analysis known to author (Harris and Silverstone (1999))

dealt with TAR and M-TAR models that use residuals from long-run regression with linear trend and impulse dummies. One must be aware that any conclusions should be based on strong results. Ideally, Monte Carlo experiments should be used to evaluate size and power of all tests as well as critical values. Mentioned article stated that properties of the series are changed but not dramatically. Authors conclude that decisive results of the tests based on the harsh critical values are unchanged.

M-TAR model were initially advised in analysis of systems in which adjustment was assumed to come from the outside. For example, consider analysis of the exchange rate in the presence of managed float. In such case the exchange rate authority may want to mitigate large changes in the exchange rate without attempting to influence the long run level of the rate.

Enders and Siklos (1999) considered the term structure of the interest rates and concluded that M-TAR model might be used to test for asymmetry in such case because the Federal Reserve might take strong measures to offset shocks to the term structure relationship when such shocks are deemed to indicate increases in inflationary expectations. Shocks that indicate decreases may not be offset so readily.

Caner and Hansen (1998) argued that M-TAR adjustment might be superior to simple TAR-type adjustment on purely statistical basis. According to their work, if $\{u_t\}$ is a near unit root process, setting the indicator using lagged first difference of the residuals (just as in (15)) can perform better than the specification described by (12) and (14) with the indicator set according to (13).

The M-TAR model is particularly useful in detecting asymmetrically “steep’ movements in a series. The TAR model may be preferred when one is interested in capturing asymmetrically “deep’ movements. For the discussion of those properties of the series see Sichel (1993).

For our purposes it is enough to say that negative “deepness’ (i.e. $|\pi_1| < |\pi_2|$ in (12), (14) with the indicator set according to (13)) of a series implies negative skewness relative to the mean or trend. The deviation of observation below this mean or trend exceeds in such case the average deviation of observation above. Of course, positive deepness suggests the opposite. “Steepness’ of the time series implies that its first differences exhibit negative skewness. In such case the sharp decreases in the series are larger but less frequent than more moderate increases.

The figure below adopted from Sichel (1993) shows clearly what mentioned definitions mean:

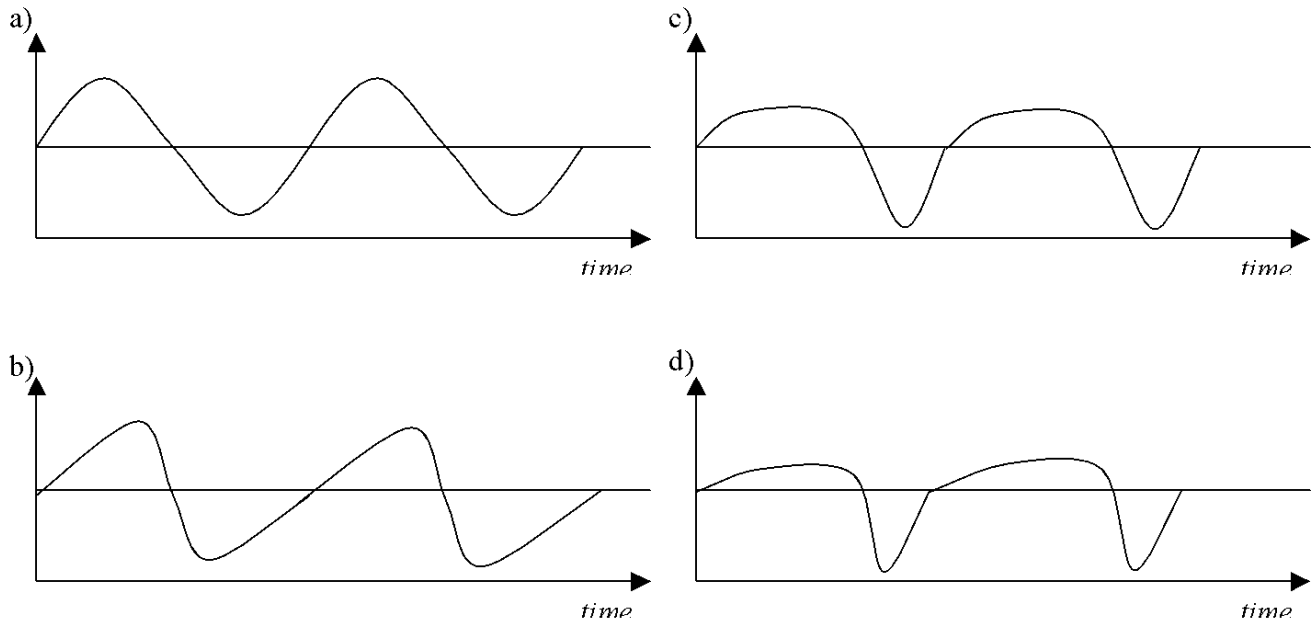

Figure 4 Characteristics of time series:

a) symmetric; b) steep; c) deep; d) steep and deep.

It is important to stress that the character of the M-TAR type asymmetry is fundamentally different from the asymmetry that is tested for by the Houck, ECM and TAR approaches. According to the M-TAR approach, a correction to the margin between prices at different levels of the transmission chain does not depend on the size of this margin at a given point in time but rather on the magnitude and direction of its *change* in the previous period. It is in this sense that M-TAR-type asymmetry is said to exhibit “momentum’.

2 An Application to UK Petrol Market - Long-run Relationship

2.1 The Production and Distribution of Petrol

The process of price transmission is a complicated mechanism. It functioning depends on the response of many agents. Understanding its nature is necessary to grasp the issues connected with the long-run equilibrium and asymmetries.

Petroleum products (petrol, diesel fuel, jet fuel, etc.) are obtained from refining crude oil. In 2001 over 90% of refinery throughput was made up from crude oil delivered from under the North Sea (Brent). Over the sample period crude oil was extracted domestically and imported (mainly from Norway) (Figure 5 shows appropriate values for period 1982-2000).

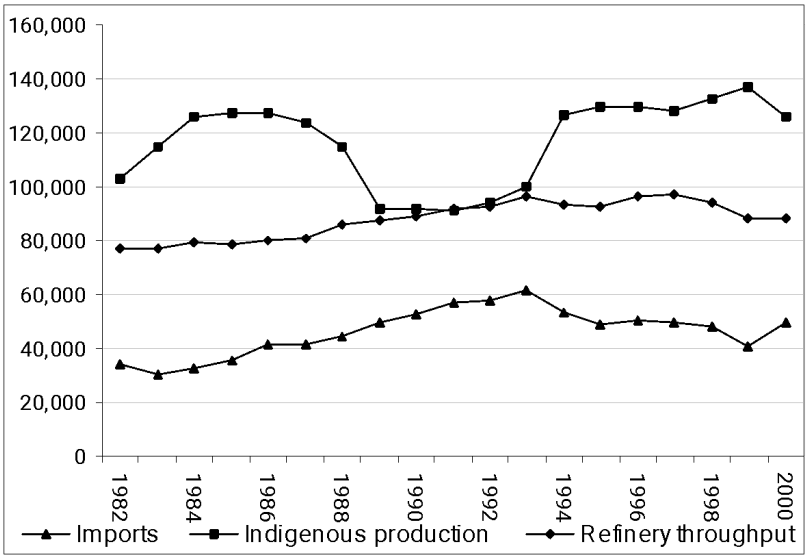

Figure 5 Crude oil: imports, indigenous production and refinery throughput ('000 tones)

Source: The DTI Energy Report (2000).

Bulk of trade in which British companies are engaged is often referred to as the "Rotterdam market'. This generic term is given to trade in oil product in Northwest Europe and takes its name from the large refining and storage complex in the Antwerp, Rotterdam and Amsterdam area. The predominance of the Rotterdam spot market in Europe has been established many times e.g. by Indjehagopian and Simon (2000).

Prices of Brent API 37.4°, the European benchmark crude used in my dissertation as a proxy for upstream prices, on "Rotterdam market' are correlated with spot prices of other crudes and indexes/composites used for determining spot and futures prices (the most popular is published daily in Platt's Oilgram News). It is natural since crude oil is traded internationally and all varieties of crude oil are close substitutes. Table 1 reports the correlation coefficients between various kinds of crude oil. Data on North Sea crude composites from April 1983 to December 2001 was obtained from the Institute of Petroleum. Data on US composites was obtained from.

Name of the crude	Correlation coefficients between levels of prices
North Sea crude composite	.97419
US OPEC Composite	.95750
Arabic Light Composite (mainly Dubai API 32°)	.92665

Table 1 Correlation coefficients between prices of crude oils

Source: Institute of Petroleum (2002), EIA (2002).

After transporting, usually by sea, the crude is processed in the refineries. Refining corporations in the UK are usually divided into two groups "majors' and "mini-majors' according to their size and presence on the global market. Figure 6 reports aggregated volumes of refinery output, inland deliveries of petroleum product and motor spirit from 1982 to 2000. Table 2 shows shares of output of majors and mini-majors in 1988 and 1996.

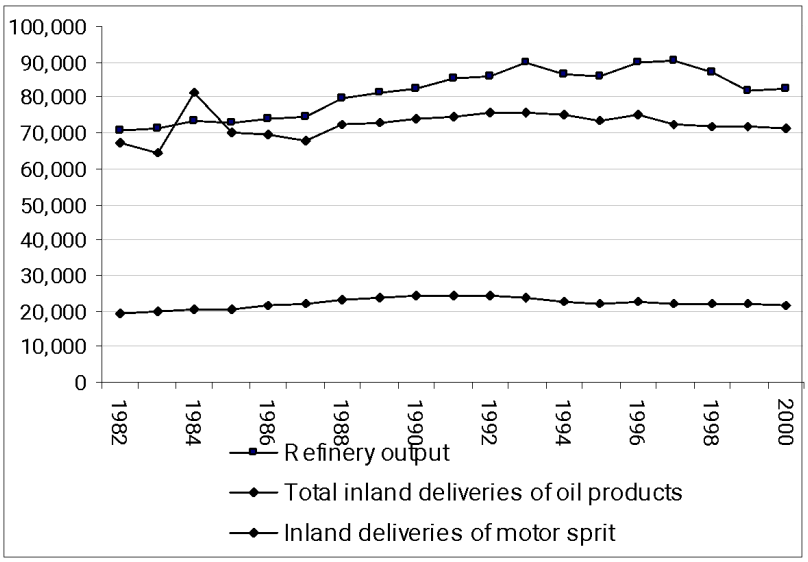

Figure 6 Refinery output, total inland deliveries of oil product and motor spirit ('000 tones)

Source: The DTI Energy Report (2000)

Status	Company	1988(%)	1996(%)
Majors	Esso	18.0	17.9
	BP	9.2	17.1
	Mobil	10.3	
	Shell	15.4	13.2
	Texaco	12.9	10.7
	Majors total	**65.8**	**58.9**
Mini-majors	Conoco	10.7	12.6
	Gulf (Chevron)	7.4	8.8
	Elf	Nil	6.4
	Fina	4.8	5.3
	Total	4.4	5.3
	Murco	1.8	2.7
	Amoco	3.3	Nil
	ICI	1.9	Nil
	Mini-majors total	**34.2**	**41.1**

Table 2 Shares of UK refined output by companies 1988 and 1996

Source: The OFT Report (1998)

Note: In late 1996 there was a merger between Mobil and BP. Data for the last 6 years was not available.

Prices paid on this stage are either directly taken from the "Rotterdam market' or are based on the indexes that account for the movements on other markets. What is natural, they are expressed in US dollars.

After the refining process, a proportion of petrol is exported or collected at the refinery gate and distributed to retail sites by a wholesaling arm of the refiner (refining wholesaler). A refinery can also supply other refiners and non-refining wholesalers.

Figures in Table 2 do not correspond directly to market share for supply to wholesaling because some of the output is exported or imported but they map quite well the structure of the industry with 4 majors accounting for almost 60% of output.

Products also can be transported by inland pipeline network, by sea or by rail to distribution terminals where they wait for other wholesalers.

Most of the prices on this stage are linked to the Platt's prices of processed products plus a small premium. Prices charged to non-refining wholesalers are similar to those charged to refining wholesalers - the average prices according to the OFT report vary by only 1.2 p per litre which reflects location of the pick-up (terminal or refinery gate) and charges for the refiner holding the associated stock obligation.

The share by volume of petrol supplied by wholesalers is given in Table 3.

Company	1988(%)	1997(%)
BP	12.6	22.0
Mobil	7.7	
Esso	19.2	20.0
Shell	16.5	19.5
Gulf	3.0	
Texaco	9.5	8.0
Conoco	6.1	6.5
Elf	3.0	4.5
Fina	3.9	4.0
Total	3.9	4.0
Burmah/Save	*4.6*	*2.5*
Kuwait	*2.3*	*2.5*
Murco	1.2	1.5
Amoco	1.8	Nil
Others	4.6	5.0

Table 3 Wholesale supply to retailers in 1988 and 1997

Source: The OFT Report (1998)

Note Entries in italics are for non-refining wholesalers, Gulf Oil (Great Britain) Ltd was acquired by Shell UK ltd on 5 XII 1997. Data for the last 5 years was not available.

Typically, wholesalers have a nation-wide baseline price from which the price charged to retailers (mainly independents) is derived. They deduct from that price an amount that can be thought of as a recommended gross retail margin ('dealers margin').

In addition to the standard agreements there exist some discretionary payments made by wholesaler in order to attract clients. They are deducted directly from the wholesale price at the time of delivery or they are paid as a lump sum in arrears. Deductions of the first kind influence the marginal benefit to the retailer of selling one extra unit of petrol and are likely to influence retailers' pricing policies. One of the most common adjustments made to the wholesale price is called SPS.

It is a condition of payment that the deduction is passed on in full to the customer in lower petrol prices. It is usually withdrawn when retailer misused it to enhance his own margin. These payments are designed to help the retail site to remain competitive with the local rivals.

Up to 1995 such support was given only to selected retailers, typically at a local level and was aimed at helping to cope with local competition. Later on, with the entry of sites owned by the supermarket chains the situation has changed.

The emergence of the supermarket-owned sites is seen as the main development in the market for petrol over the last 20 years. They started to sale petrol in the late 1980s and quickly gained the market share becoming serious threats for established companies. Figure 7 shows developments in the number of retail sites in the 1990s. Increase in the number of supermarket-owned sites and decrease in the number of independent sites are the most visible trends.

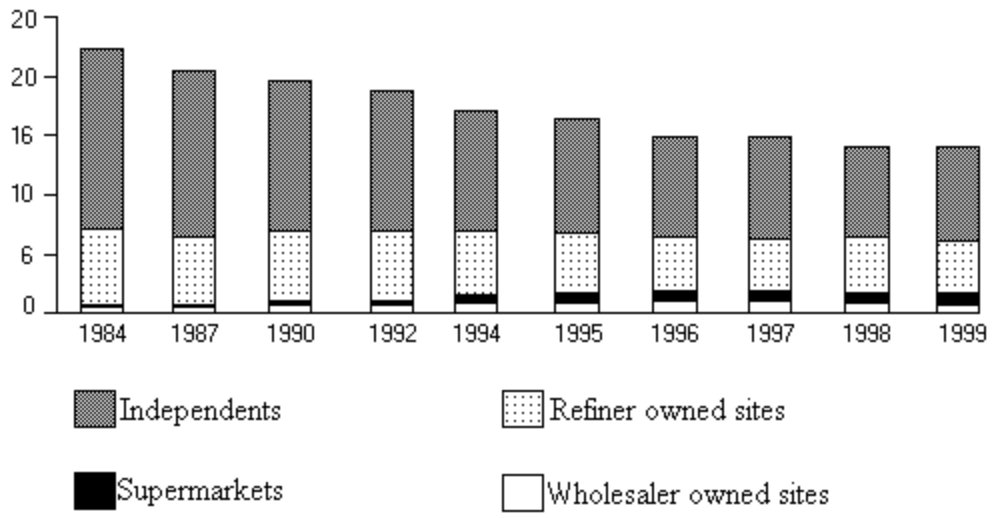

Figure 7 Number of retail sites in the UK ('000)

Source: Williamson and Taylor (2000:5)

Note: In 1997 a revised methodology was introduced which resulted in the inclusion of 871 sites previously missing from the survey, hence the apparent rise in numbers between 1996 and 1997.

The advantage the supermarkets enjoyed was possible because they were close to busy shopping centres, they focused on the price of wholesale petrol regardless of its origin and they had huge throughputs allowing for exploiting the economies of scale.

For example, according to the OFT (1998:31) average site annual throughputs in 1996 were (in million litres): 8.59 for supermarkets, 2.56 for company owned sites, 1.16 for dealer owned sites. Figure 8 shows the evolution of the supermarkets' share of UK deliveries of motor spirit in the 1990s.

Figure 8 Supermarkets' share of the volume of UK motor spirit retail deliveries (%)

Source: Williamson and Taylor (2000:5).

In the mid-1990s the loss of volume to supermarkets prompted the major oil companies into severe price reactions. Late in the 1994 BP announced that it will do its best to price nationally in response to the supermarkets but it was in January 1996 when policies followed advertising campaigns. Then, shortly after launching its Pricewatch campaign in Scotland and North East England Esso rolled out Pricewatch on a national basis. This was soon followed by Shell's declaration that it will match Esso prices everywhere. The OFT report (1998: 61) described subsequent events as

> ' ... an intense seven-month price war (...) during which approximately 10% of service stations in the UK closed'.

The aim of this campaign was to match the supermarkets by offering SPS to retailers so as to enable them to offer the lowest prices. The strength of this programme came from the fact that it did not require any contribution from the retailers - all the costs were borne by the majors. For dealer-owned sites Pricewatch required forgoing some of the dealer margin and decreased their profitability for refinery-owned sites it was less costly. First months of the campaign were marked with closures of the sites and sudden decreases in the retail prices. Later on, when supermarkets' advantage was not so large, majors and mini-majors returned to their normal practices.

2.2 Data Used, Properties of the Series, Tests for Stationarity

I represented the complex chain of transformation of crude oil into 4 Star petrol described in Chapter 2.1 as a one-stage process. I consider together transformation of oil into refined products and its distribution to petrol stations. The main drawback of such representation is the constraint imposed on the analysis. In my approach, it is impossible to attribute asymmetry to a particular stage in the transmission chain and distinguish between sources of asymmetry with confidence.

My approach is consistent with that of Sumner (1990), Manning (1991), Reilly and Witt (1998). What is more, other analyses (Borenstein et al. (1997), Godby et al. (2000)) also dealt with transmission from crude oil prices directly to retail prices, treating it as an introduction to a more sophisticated analysis.

Following Manning (1991), I considered the crude oil to be the only significant cost to affect the price paid by drivers. Data on costs of transportation, storing crude oil and petrol was not available on any reliable and constant basis over the sample period.

The frequency of the data also requires short explanation. Semi-monthly or weekly data are superior to monthly data used in this dissertation. Data on more frequent basis would allow me for analysis of cyclic patterns and distinguishing between more sophisticated explanations of asymmetry such as price cycles. But again, in the bulk of the analysis of U.K market monthly data are used, as it is difficult to obtain any data on different basis.

I decided to focus my attention on leaded petrol. Analysis of prices of unleaded petrol would be constrained to the last 10-12 years. My sample covers last 20 years (January 1982 to December 2001) and it seems that it is the longest time period ever examined for the UK petrol market.

The variables in my analysis are the monthly series for retail UK petrol price (p) net of VAT tax and the excise duty, Brent oil price (c) and the dollar/sterling exchange rate (ex).

Data on typical retail prices of leaded gasoline classified according to the British Standard BS4040 as 4 Star (i.e. 97 octanes) was taken from the Digest of UK Energy Statistics. The data is representative of prices paid on or about the 15th of the month and is based on information provided by oil companies. From January 1995 data includes supermarket chains. Effective rates of duty on principal hydrocarbon oils were also obtained from DTI, details about the rules governing them were obtained from the Institute of Petroleum.

From October 1999 4 Star prices published by DTI represent also “Lead Replacement Petrol (LRP)’. The excise duty levied since March 2001 depends upon the sulphur and aromatic content of the fuel. Duty on environment-friendly fuel is lower. Not all leaded petrol can be classified in this way. Official statistics since March till December can incorporate prices of 2 kinds of leaded petrol subject to two different duties (see Smith (2000) for details). This may affect the properties of the data. Further details will be provided later in this chapter.

Prices of crude oil are obtained from DataStream. They represent average FOB spot prices in USD paid at the beginning of the month for one barrel of Brent API 37.4° crude oil on so-called “Rotterdam market’.

The exchange rate is introduced because the prices of crude oil are expressed in US dollars. As pointed out by Beacon (1991), Reilly and Witt (1998), Galeotti et al. (2002) the exchange rate may be relevant source of asymmetry in non-US countries. Hence the importance of allowing separately for positive and negative changes in exchange rates. Average monthly exchange rate between US dollar and British pound was taken from DataStream.

Later, when apparatus used is not sophisticated enough to allow for separate treatment of the exchange rate, the price of crude oil will be expressed in pounds.

Simple logarithmic transformation was applied to all the series in order to prevent problems caused by quadratic trends (Banerjee et al. (1993)). Borenstein et al. (1997) pointed out that using logarithms implies that crude-retail margin increases with the price of crude oil. As an alternative he proposed using the raw data. Unfortunately, that implies constant nominal margin in the level equation, which also can be problematic and is not consistent with the findings of the OFT (1998) (chapters 7-9).

What is more, Borenstein et al. (1997: 312) did admit that with longer samples, which include observations from periods of high and variable inflation, their approach would not be justifiable.

Over the whole sample period the properties of the series were investigated using the typical array of tests.

To avoid bias caused by exclusion of the relevant trend in the data generating process the non-standard F-test (Reilly (2002)) for the null hypothesis of root in the unit circle and no deterministic trend was applied for all series. In all cases the statistics were smaller than the

critical values for 250 observations and 5% significance i.e. 6.34 (for logs of retail prices, crude oil prices and the exchange rate the results are, 3.7397, 4.9349 and 2.7350).

In case of crude oil prices and exchange rate autocorrelation was present. To whiten the residuals I included lags of dependent variable. In case of exchange rate introducing only one lagged variable ($\Delta ln(ex)_{t-1}$) increased all information criteria and resulted in increased F-test statistics (4.2187). The same was the case for crude oil prices. Therefore in testing procedures I included tests that take into consideration presence of the deterministic trend.

Augmented Dickey-Fuller tests with lag order of twelve in all cases lead to conclusion that unit root may be present in all series. Exact lag length for each test was taken in accordance to the highest Information Criteria (IC). When any doubts were encountered HQC and the SBC were treated as superior to AIC since in the latter the penalty function is asymptotically less harsh.

For all series rejecting the null of a unit root at 1% significance level was impossible. For retail prices the test statistics exceeded the critical values at 5% level. Further computations revealed that excluding only 4-5 last observations, when 2 different duties were levied, gave opposite results. Plots of autocorrelation functions for all level variables pointed out slow decay (20 months) characteristic of first-difference stationary series.

Alternative tests - CDRW for residuals from regressing on a constant were also performed, they supported the hypotheses of all series being I(1).

With usual care, series could be classified as first-difference stationary. Although in some cases it required using critical values harsher than 5%, classifying them in such way, may have prevented problems with estimation the level equations (like those encountered by Balke et al. (1998)).

To ensure that the prices are indeed I(1), the same array of diagnostic tests was applied to first differences. In all cases null hypothesis of a unit root was rejected at typical significance levels. Results are presented in Table 4.

All entries represent absolute values of the statistics. Bold numbers when null of series being I(1) can be rejected. Appropriate CV at 5% significance level: intercept and no trend for longer sample -2.8743, intercept and trend for longer sample -3.4307, intercept and no trend for shorter sample -2.8745, intercept and trend for shorter sample -3.4310.

	CRDW	Dickey-Fuller tests

Test series		With intercept without trend					With intercept and trend				
		DF	ADF	ADF	ADF	IC	DF	ADF	ADF	ADF	IC
Lag length	N/A	0	6	12	Recommended		0	6	12	Recommended	
ln(p) (till 2002)	.1203	2.650	2.475	**3.016**	**3.104**	1 all criteria	2.640	2.452	3.019	3.099	1 all criteria
ln(p) (till Sep 2001)	.1189	2.472	2.361	2.744	**2.910**	1 all criteria	2.425	2.293	2.679	2.870	1 all criteria
ln(c)	.1252	**3.1453**	2.756	**3.192**	**3.145**	0 SBC HQC	3.111	2.658	3.173	3.111	0 SBC HQC
ln(ex)	.0664	1.8705	2.309	2.514	2.652 2.622	1 SBC 3 HQC AIC	1.826	2.278	2.483	2.632 2.597	1 SBC 3 AIC HQC
Δln(p)	**1.7166**	**12.933**	**6.285**	**4.371**	**12.933**	0 all criteria	N/A				
Δln(c)	**1.8534**	**14.002**	**6.655**	**4.376**	**14.002**	0 all criteria					
Δln(ex)	**1.4014**	**11.054**	**6.3108**	**4.484**	**10.508** **7.557**	1 SBC 2 HQC AIC					

Table 4 Test statistics for all series of interest.

Findings about the process governing the series are consistent with the bulk of the literature except for Balke et al. (1998), who found that their series were stationary.

2.3 Long-run Relationship, Testing for Cointegration

Application of OLS to the level equation with I(1) series may lead to "nonsense' or "spurious' regressions unless variables are cointegrated. This issue was considered so many times that any details might be omitted (for a review of the literature and theoretical background see Banerjee et al. (1993) or Maddala and Kim (1998).

Following Sumner (1990), Manning (1991), Shin (1994), Reilly and Witt (1996), Borenstein et al. (1997) I assumed that crude oil prices and the exchange rate are exogenous. That implies the direction of the causality: from prices in North West Europe to retail prices in the UK. It also implies that shocks to retail prices in the UK do not affect prices of crude oil in the North West Europe and the exchange rate.

I estimated simple linear model with and without deterministic trend and assumed linear relationship between logs of net retail petrol prices, the exchange rate and crude oil prices.

Estimation of the level equation in the form given by (2) lead to following results[5]:

$$\ln(p)_t = \beta_1 + \beta_2 * trend + \beta_3 * \ln(c)_t + \beta_4 * \ln(ex)_t + \varepsilon_t$$
$$\ln(p)_t = 1.3894 + .003729 * \text{trend} + .49011 * \ln(c)_t + .37999 * \ln(ex)_t \qquad (16)$$

$$\ln(p)_t = \beta_1 + \beta_2 * \ln(c)_t + \beta_3 * \ln(ex)_t + \varepsilon_t$$
$$\ln(p)_t = 1.5126 + .46407 * \ln(c)_t + .37947 * \ln(ex)_t \qquad (17)$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto-correlation[6]	RESET	ARCH	Hetero-sceda-sicity	JB Normality
Model without a trend	Value	.68	.36	210.68	205.46	163.3	1.305	111.78	3.20	.618
	P-value					.000	.253	.000	.074	.734
Model with linear trend	Value	.70	.40	217.33	210.37	157.2	.58	122.3	1.07	11.56
	P-value					.000	.446	.000	.300	.003

[5] In all estimation results the t-test statistics are reported in square parentheses. For I(1) variables the t ratios are omitted.

[6] In all tests labelled auto-correlation in my dissertation the number of degrees of freedom (d.o.f.) equals 12, RESET tests for specification are with 1 d.o.f., Arch tests are with 12 d.o.f. , tests for heteroscedascity are with 1 d.o.f.

On the basis of test statistics I concluded that model with trend is superior to that without. One additional advantage of the linear trend is that it can capture effects of other costs on the retail petrol prices (for example real wages were increasing over the sample period).

Results from estimations of (16) and (17) do not differ substantially. The only problem caused by introducing the deterministic trend is a normality of residual as shown by JB statistic.

As it will be shown in Chapter 3.2 introduction of only 5 impulse dummies to control for the beginning of the Pricewatch campaign resulted in different tests statistics indicating normality of the residuals.

As a first test for cointegration the ADF test was applied to residuals obtained from estimation as in (11). Again the optimal lag length was chosen in accordance with information criteria. SBC, HQC and AIC criteria indicate that typical DF statistic without lags is appropriate. Its value -5.1729 exceeds the critical value at 5% significance level -4.1725 allowing us for rejecting the null of residuals being I(1). Also CRDW statistic exceeds critical value for 4 regressors i.e. 0.3. indicating cointegration.

The critical values based on response surfaces also support the view that the long run relationship between prices exist. Regressing changes in of residuals on residuals lagged one period as in (11) gave t-statistic equal to -5.1514. The critical values at 1% and 5% significance level are -4.74551, -4.16963, accordingly. Restricted and unrestricted ECMs were also used to test for cointegration. Using lagged residuals from (16), equation (6) was estimated:

$$\Delta \ln(p) = \alpha + \beta_1 * \Delta \ln(c)_t + \beta_2 * \Delta \ln(ex)_t + \pi * \hat{u}_{t-1} + \varepsilon_t$$

$$\Delta \ln(p) = .033287 + .27464 * \Delta \ln(c)_t + .20636 * \Delta \ln(ex)_t - .18176 * \hat{u}_{t-1} \qquad (18)$$

[.094463][8.1126] [1.5272] [4.9983]

							Lagrange Multiplier χ^2 tests			
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-sceda scity	JB Normality
(18)	Value	.268	1.89	357.73	350.78	16.36	.125	7.12	2.17	81.88
	P-value					.175	.724	.856	.145	.0

The model above passes the array of diagnostic tests. The fact that residuals do not pass the JB test does not affect the efficiency of the OLS estimation. The t-statistic on the lagged

residuals suggests that they are significantly different from zero, taking into consideration the critical values at 5% and 1% significance level as given by McKinnon (1991). It also supports the hypothesis that all series are cointegrated.

As the next step in testing for cointegration unrestricted ECM in the form given by (7) was employed.

$$\begin{aligned}
\Delta \ln(p)_t &= \alpha + \beta_1 * \Delta \ln(c)_t + \beta_2 * \Delta \ln(ex)_t \\
&\quad + \pi_1 * \ln(p)_{t-1} + \pi_2 * \ln(c)_{t-1} + \pi_3 * \ln(ex)_{t-1} + \varepsilon_t \\
\Delta \ln(p)_t &= .24115 + .27560 * \Delta \ln(c)_t + .20638 * \Delta \ln(ex)_t \\
&\quad [.3.4946] \quad [7.9263] \quad [1.4856] \\
&\quad - .16974 * \ln(p)_{t-1} + .084461 * \ln(c)_{t-1} + .067155 * \ln(ex)_{t-1} \\
&\quad [\text{N/A}] \quad [\text{N/A}] \quad [\text{N/A}]
\end{aligned} \tag{19}$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-sceda scity	JB Normality
(19)	Value	.263	1.90	354.93	344.5	17.54	.375	7.63	1.66	84.69
	P-value					.130	.540	.813	.197	.000

In this framework testing for cointegration involves performing non-standard F-tests (so-called Peseran test) for join insignificance of all lagged values (Peseran, Shin and Smith (1996)). In the model above it equals to 7.6811 and exceed the critical value for 3 variables, intercept and a trend for 5% and 1% significance level (i.e. 5.972 and 7.584, respectively).

All tests above indicate that long-run relationship between variables under scrutiny really exists. This rules out possibility of "spurious regression'. Since it was proven that with cointegration OLS estimators are not only consistent but also N consistent, we can treat βs from estimations above as reliable estimates of real parameters.

The point is that since those estimates are superconsistent OLS estimation gives residuals that represent unpredictable shocks to the system. This gives base for several of the tests for asymmetric adjustment that will be used in Chapter 3.

One remaining problem is connected with the stability of the parameters. To account for it the CUSUM and CUSUMSQ tests were applied to the whole sample (for a discussion on those tests see (Brown, Durbin and Evans (1975)).

As the picture below shows the null hypothesis of the parameter stability over the sample period must be rejected at 5% significance level.

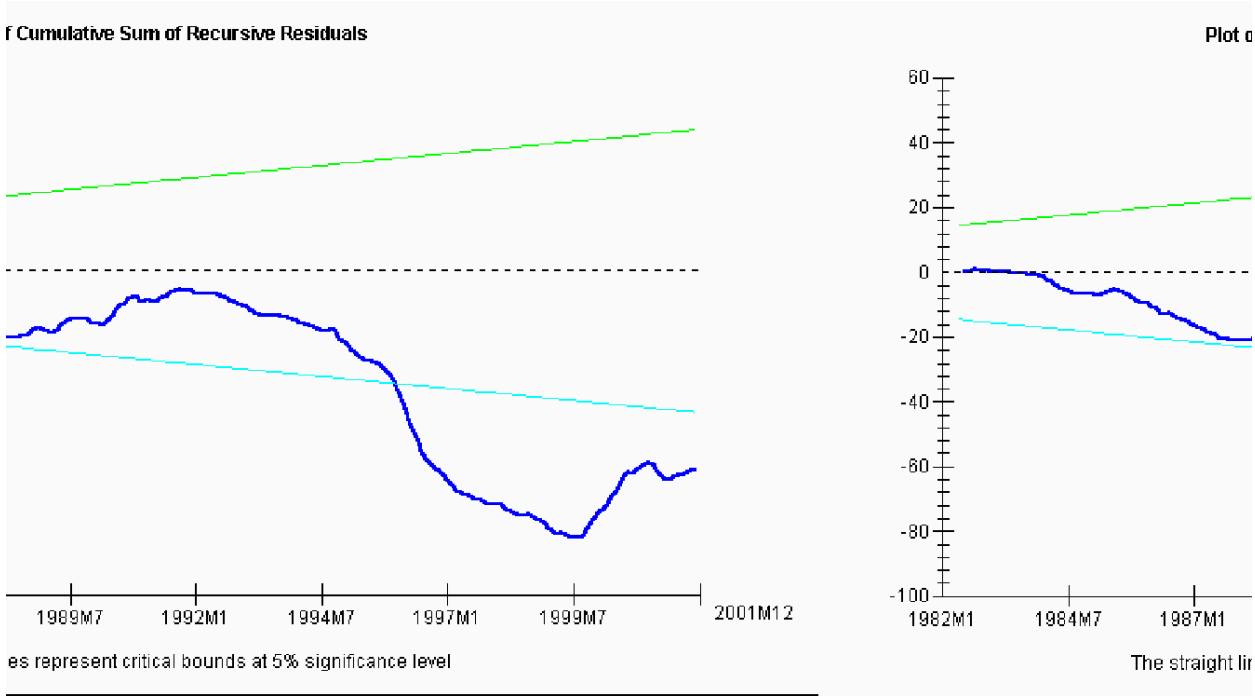

Figure 9 CUSUM test

Straight lines represent the critical values at 5%

As the next step to ensure if the changes were caused by one-time deviation or change in the parameters the CUSUMSQ test was applied. Result given below suggests instability in the long-run relationship around 1996.

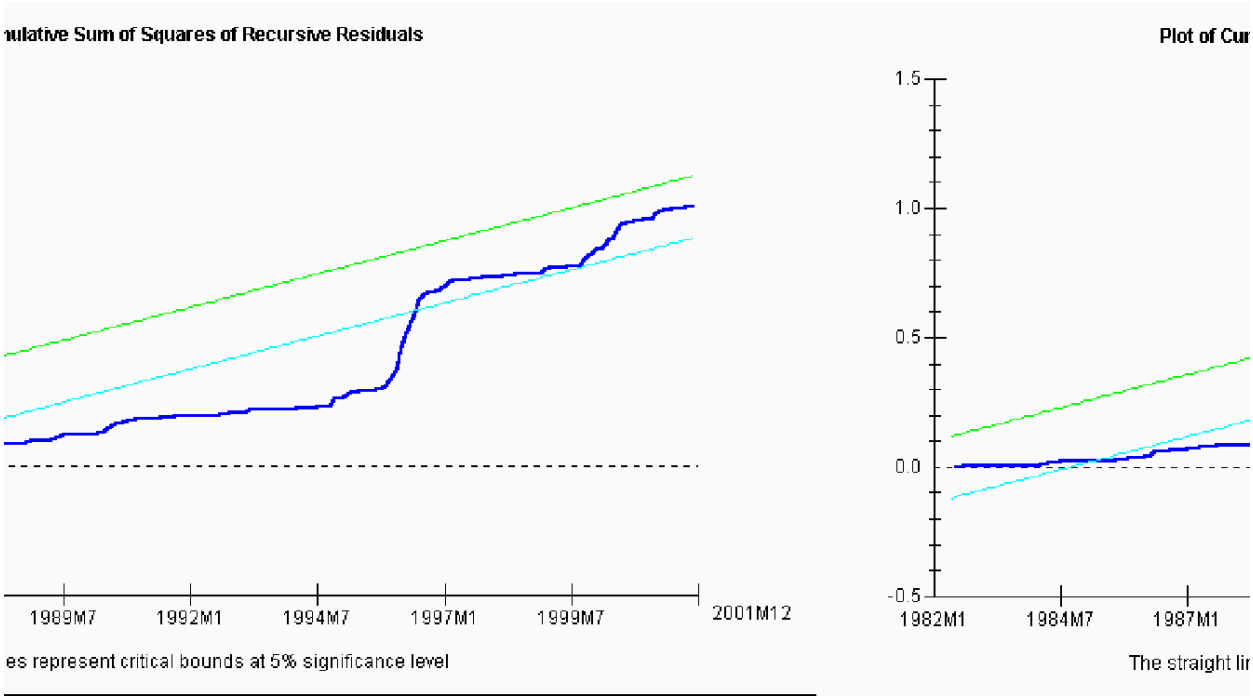

Figure 10 CUSUMSQ test

Straight lines represent the critical values at 5%

Also the behaviour of the standard errors of recursive regression was examined. As Figure 11 shows, around 1996 there is a sudden increase in the value of standard error.

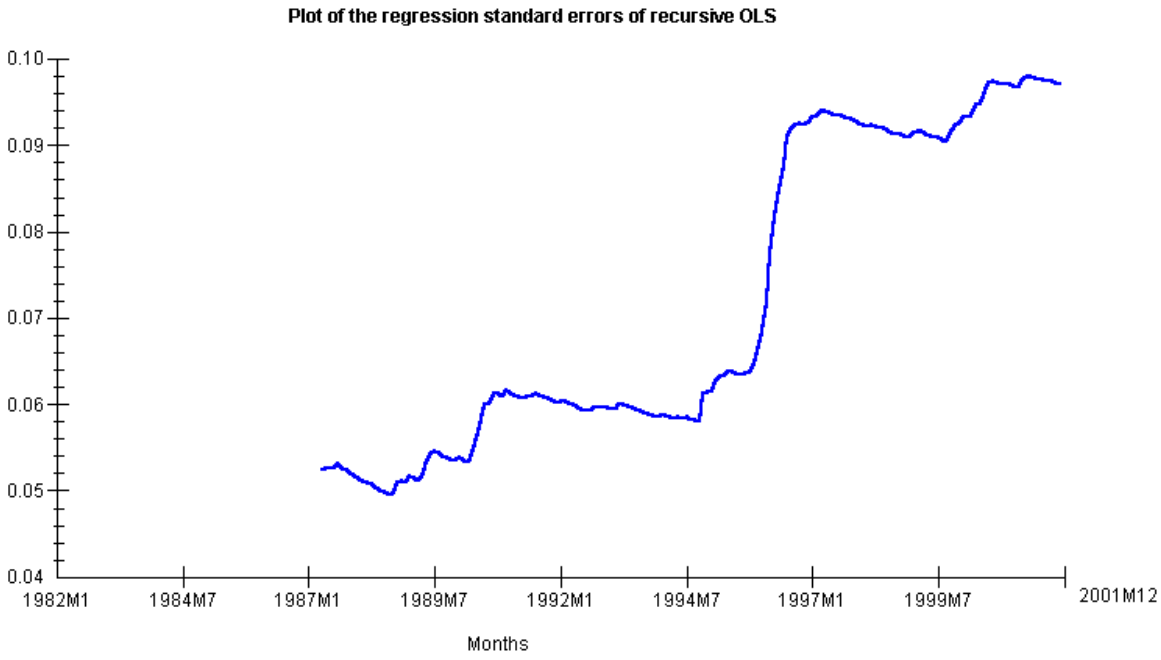

Figure 11 Standard errors of recursive regression.

To check thoroughly the properties of (16) I performed one-step-ahead Chow test. For calibration of the model I used the first 10 years and the standard error for that period (.058551). The test statistic scaled by the 5% critical value for the χ^2 distribution (3.84) in Figure 12 is represented by the blue line.

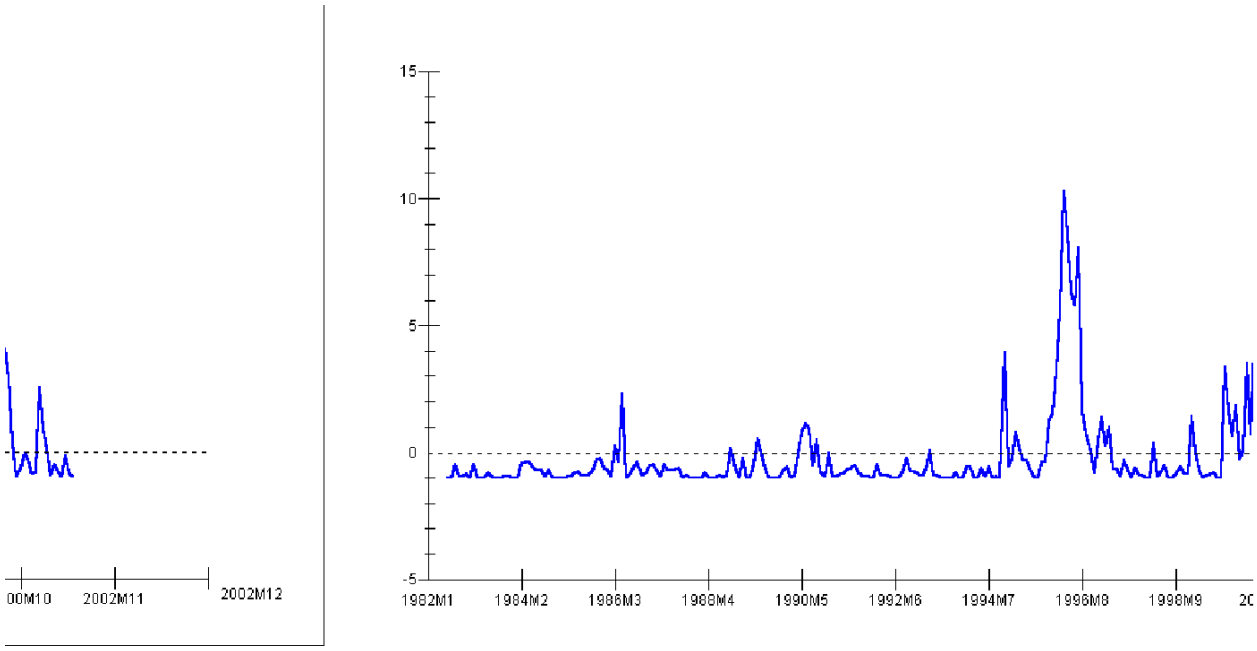

Figure 12 One-step-ahead Chow test

I conclude that the relationship cannot be seen as perfectly stable over the whole sample. The most probable explanations are connected with the Pricewatch programme.

2.4 Conclusions

In this chapter I established the existence of the long-run relationship between variables in question. Analysis of their statistical properties suggest that they are first-difference stationary so if they do cointegrate OLS estimators are N consistent. Therefore one can treat them as reliable estimates of the true parameters. If this is the case, the shocks to the system will be reflected in the values of the residuals. This gives the basis for the use of threshold autoregressive models. What is more, techniques based on ECM can also be used. Findings from this chapter allow for application of techniques necessary to analyse asymmetry.

3 Testing for Asymmetry

3.1 Asymmetry in the Short-run Adjustment

To test whether the short-run response to changes in the crude oil prices is symmetric the ECM model as described by equation (8) was estimated over the whole sample. Following Banerjee et al. (1993), the Hendry's 'general-to-specific' procedure was used to determine the appropriate lag structure. At each step of the procedure individual coefficients on first differences *(βs)* were excluded if they were not significant at 0.5%, according to following rule:

$$1-\prod_{i=1}^{10}(1-0.005)=0.048889 \tag{20}$$

Moreover, the tests for the join insignificance of excluded variables were also performed. For testing the null hypothesis of symmetry in short-run response conventional critical values were used i.e. the t-statistics on the coefficient on positive first differences *(β^+s)* were compared to conventional critical values at 5% significance level. After the testing down the following model was obtained:

$$\begin{aligned}\Delta\ln(p)_t &= \alpha_1+\alpha_2*trend+\beta^+{}_{1,0}*\Delta^+\ln(c)_t+\beta_{1,0}*\Delta\ln(c)_t+\beta_{1,1}*\Delta\ln(c)_{t-1}+\\ &\beta^+{}_{2,0}*\Delta^+\ln(ex)_t+\beta_{2,1}*\Delta\ln(ex)_{t-1}+\pi_1*\ln(p)_{t-1}+\pi_2*\ln(ex)_{t-1}+\pi_3*\ln(c)_{t-1}+\varepsilon_t\end{aligned} \tag{21}$$

The deterministic trend was introduced following Reilly and Witt (1998) to account for omitted variables such as labour cost. Introducing the trend is common in the literature (e.g. Borenstein and Shephard (1996)) and this study is no exception. Author decided for it in order to compare the results with previous studies on the UK market.

After further analysis several restriction on the coefficients were imposed. The restriction that the coefficients on $\Delta ln(c)_t$ and $\Delta ln(c)_{t-1}$ are equal was imposed after applying the Wald test, which gave test statistic equal to 1.7877 (lower than the critical value at 5%). Also the coefficients *on* $ln(c)_{t-1}$ and $ln(ex)_{t-1}$ were equal to each other – it was proved by the Wald test statistic equal to 0.712 with p-value equal to 0.777. Test for join insignificance of all variables excluded in the 'general-to-specific' procedure confirmed that they are not significantly different from zero.

The final model is given below:

$$\Delta \ln(p)_t = .23209 + .0002869*trend + .16086*\Delta^+ \ln(c)_t + .16357*(\Delta \ln(c)_t + \Delta \ln(c)_{t-1}) +$$
$$[3.9333] \quad [.54526] \quad [2.2934] \quad [5.7391]$$
$$.42302*\Delta^+ \ln(ex)_{t-1} + .34391*\Delta \ln(ex)_{t-1} - .15244*\ln(p)_{t-1} + .067310*(\ln(ex)_{t-1} + \ln(c)_{t-1}) \qquad (22)$$
$$[1.9265] \quad [2.5067] \quad [N/A] \quad [N/A]$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedasticity	JB Normality
(22)	Value	.341	1.94	364.87	350.98	14.809	.0665	2.3686	.135	174.05
	P-value					.252	.796	.999	.712	.000

The contemporaneous change in the exchange rate ($\Delta ln(ex)_t$) was excluded because it was completely insignificant. However, the positive first difference was significant judging by standard critical values at 5% significance. This may suggest that passthrough of favourable exchange rate may be very weak. But also the harsh criteria used in determining the optimal lag structure may be responsible. Models that include this variable are not significantly different from the one described above.

According to the results from (22) 1% increase in the prices of crude oil at *t* causes the prices of petrol to increase by 0.160%+0.163% at *t* and by further 0.163% in the next period. This is consistent with the estimate of 0.224% by Reilly and Witt (1998) and that by Galeotti et al. (2002) -.198.% Sumner's (1990) estimate of the immediate response to 1% increase in crude oil prices equal to 0.38% is quite close to my results. In contrast to that, 1% decrease in the prices of crude oil causes fall by only 0.16% at *t* and by similar amount in the next period (ceteris paribus).

The movements in the exchange rate also seem to cause asymmetric short-run response. The fact that the coefficient on $\Delta log(ex)_t$ was not significantly different from zero at 0.5% suggests that favourable change in the exchange rate is not passed to prices of petrol immediately. In the next period after 1% favourable change prices of petrol fall by approximately 0.34% (ceteris paribus). The situation with unfavourable changes is quite different – at *t* there is a transmission of 1% change equal to 0.42% and it is also present in the next period.

Galeotti et al (2002) also found that short-run response to a favourable change in the exchange rate is insignificantly different from zero. Their estimate of response to unfavourable exchange rate is equal to 0.732 and is close to that obtained by Reilly and Witt (1998). My estimate is lower, but also significantly different from zero.

It is important to add that the results above may not represent true dynamics of the system. Hendry's approach requires using very harsh significance levels at each stage of the testing procedure and it may result in omitting some dynamics.

For example, I was forced to exclude the positive first difference in the crude oil price lagged by one period. The coefficient on it was negative. This can be seen as a sign that in the next month *after* the change in the crude oil prices the adjustment is actually *greater* for decreases than for increases. This makes intuitive sense but could not have been accounted for when using Hendry's approach.

The estimates on the levels' variables in (22) can also be used to numerically solve for the long-run relationship. In a long-run steady-state, the short-run effects go to zero, ($\forall\ i,j \in N$ $ln(p)_{t-i} = ln(p)_{t-j}$ and $ln(ex)_{t-j} = ln(ex)_{t-i}$)and it is possible to re-express (22) and solve for $ln(p)_t$ to obtain:

$$\ln(p)_t = 1.5225 + .001882052 * trend + 0.441551 * (\ln(ex)_t + \ln(c)_t) \tag{23}$$

It may be important to add that in (24) the deterministic trend is not introduced to account for rising margins but for other costs, such as labour costs (see Reilly and Witt (1998) or Sumner (1991) for details). The long-run estimate reflecting the effect of crude oil prices expressed in pounds is close to the estimate of 0.58 reported by Reilly and Witt (1998). The long-run coefficient reported by Manning (1991) equal to 0.3055 is much lower than my result. The long-run estimate of unity reported in Bacon (1991) is entirely not in agreement with my findings. The Wald test of such null hypothesis supported this view with all force – null of full passthrough was definitely rejected.

As a visualisation of the changes in the values of coefficients over the sample period I report plots from recursive regression coefficients on $\Delta^+ ln(p)_t$ and $\Delta^+ ln(ex)_t$. The recursive calculated t-ratios suggested statistical significance at conventional levels over the 1990s. From late 1991 (the Gulf war) till beginning of the Pricewatch campaign, the estimated asymmetric effect for crude oil appears to be relatively stable (around 0.2). Later there is a little evidence for

a stable coefficient and the broad pattern may suggest rising asymmetric effect, maybe back to the previous level.

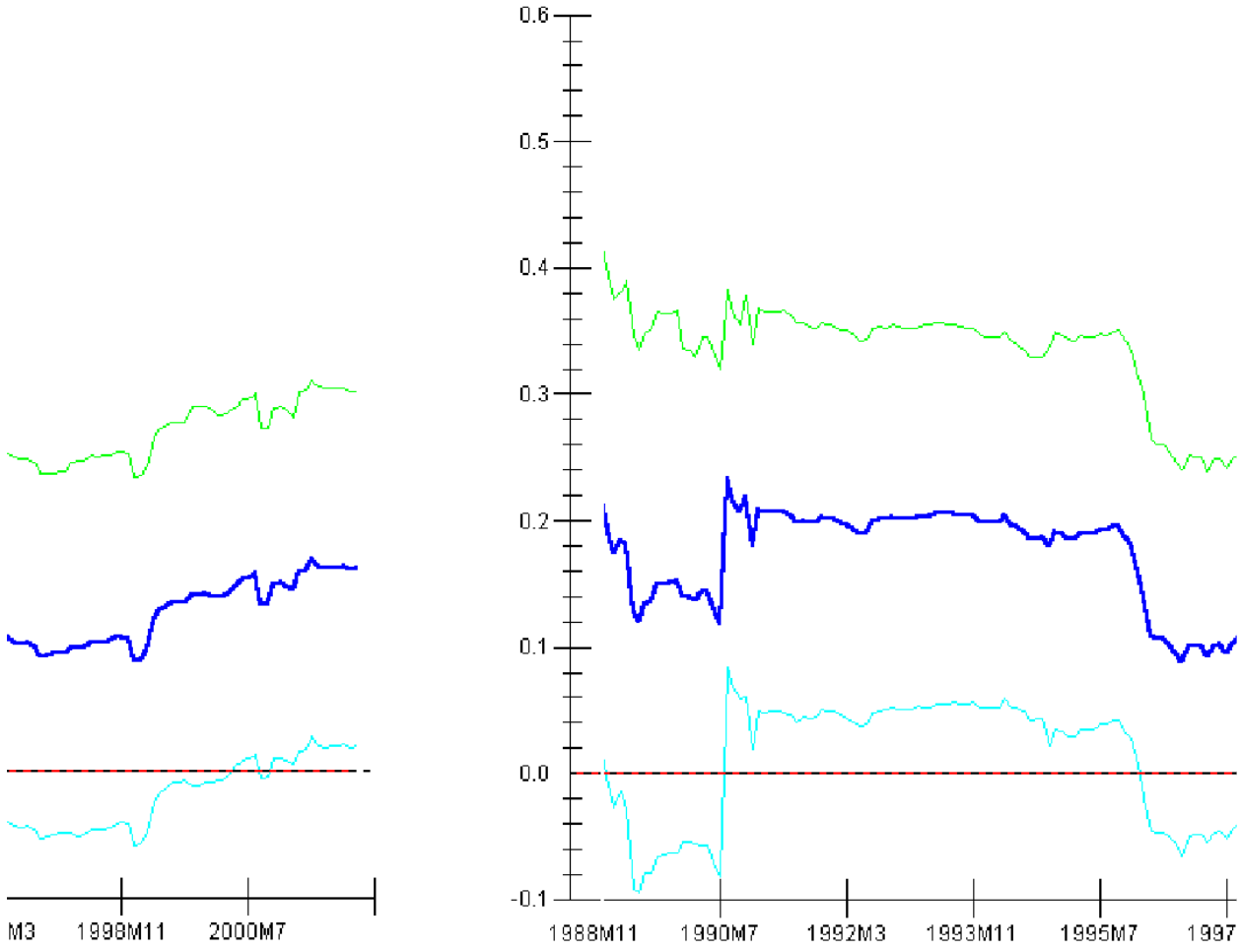

Figure 13 Plot of recursive coefficients and its standard errors for asymmetry caused by the crude oil prices

The interesting point is that results of recursive regression reported by Reilly and Witt (1998) show the same pattern with sudden increase at the beginning of the Gulf war and fairly stable values till 1995. It is worth noticing that Reilly and Witt did not report neither the t-ratios nor the standards error so as to assess the significance of the asymmetry coefficients. The figure above clearly shows that in some years the t-tests do not allow for rejecting the null of symmetric responses at the typical significance levels.

Situation with the exchange rate asymmetry seems to be slightly different. Impact of the Pricewatch campaign is not visible. The value of the coefficients seems to be fairly stable over the late 1990s.

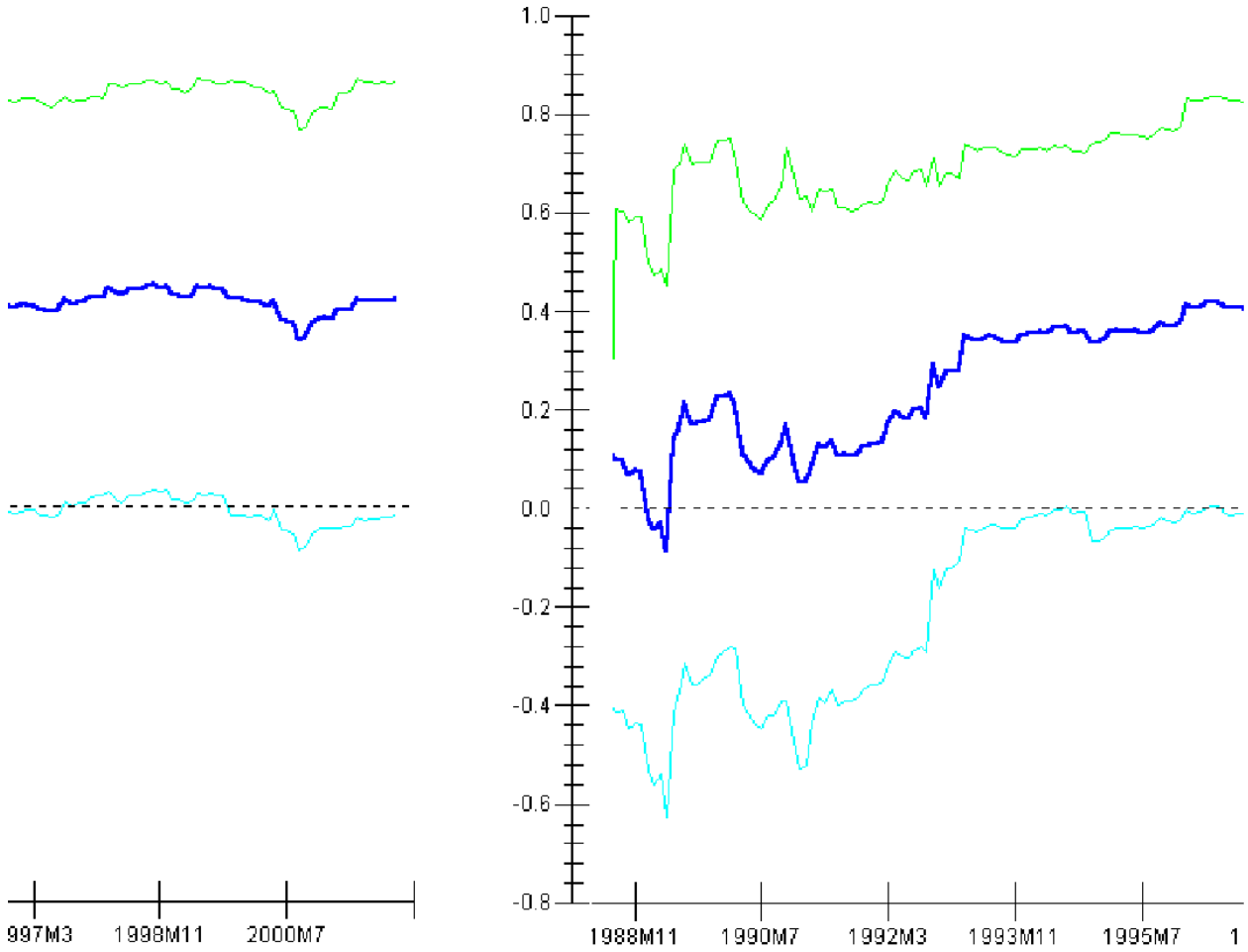

Figure 14 Plot of recursive coefficients for asymmetry caused by the exchange rate

Rising asymmetric effect visible in the early 1990s is consistent with Reilly and Witt (1998). Lack of any signs of the Pricewatch may suggest that asymmetry caused by movements in the exchange rate (i.e. occurring mainly at the refiners' level) is fundamentally different from that caused by the movements in the crude oil prices. Also in this case one can conclude that the degree of asymmetry was changing with time.

3.2 Testing for Asymmetry in the Long-run Speed of Adjustment

3.2.1 ECM Representation

Residuals obtained from the long-run equation given by (16) were used in the Error Correction Representation based on Wolfram-type segmentation as described by equation (9)

$$\Delta \ln(p)_t = \alpha + \beta_1 * \Delta \ln(c)_t + \beta_2 * \Delta \ln(ex)_t + \pi^+ * \hat{u}^+_{t-1} + \pi^- * \hat{u}^-_{t-1} + \varepsilon_t$$

$$\Delta \ln(p)_t = \underset{[.050967]}{.0002885} + \underset{[8.0640]}{.27461} * \Delta \ln(c)_t + \underset{[1.5206]}{.20645} * \Delta \ln(ex)_t - \underset{[2.4373]}{.18117} * \hat{u}^+_{t-1} - \underset{[2.8323]}{.18224} * \hat{u}^-_{t-1} \qquad (24)$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedasticity	JB Normality
(24)	Value	.268	1.88	356.73	348.04	16.623	.1350	7.1	2.18	81.9
	P-value					.164	.713	.851	.140	.000

Results from (24) show that the conditions set in Granger and Lee (1998) are satisfied, the same with those set by Cook et al. (1998).

For testing asymmetric price transmission in such setting the simple F-test on the equality of coefficient on lagged positive and negative residual is commonly used. In the setting above the null of symmetric transmission could not be rejected at typical significance levels.

However, one-step ahead Chow test, CUSUM and CUSUMSQ test, rolling and recursive regressions indicated that model above has some serious drawbacks. I attributed them to the Pricewatch campaign and I repeated all calculations taking into account possibility that the price war in 1996 could be so severe that it could change the results. To do so, for the first time in my dissertation, I used impulse dummies

In all quantitative analysis introducing any variables should be done with care. If the Pricewatch policies represent deep changes in the market not just short price war instead of impulse dummies, the slope or shift dummies should be introduced.

It is possible to analyse this problem in relation to the speed of adjustment by applying methodology described in Chapter 1.2 and using techniques aimed at looking for the changes in the values of the parameters.

I decided to apply the rolling OLS with the window size equal to 5 years to the whole sample. The rolling regression in this case was aimed at looking for the permanent and transitory changes in the speed of adjustment in the 1990s. I proceeded as follows

1) I performed ordinary rolling regression in the following form:

$ln(p)_t=\alpha_0+\alpha_1*trend+\beta_1*ln(c)_t+\beta_2*ln(ex)+\varepsilon_t;$

2) I saved the rolling coefficients and used them to compute rolling residuals, which can be seen as crude proxies for the shocks to the system;
3) Using the same window size I performed another rolling regression of changes in the residuals on lagged residuals divided into positive and negative parts as in Enders and Siklos (1999)[7]. The figures below show the values of the rolling coefficients obtained in this way.

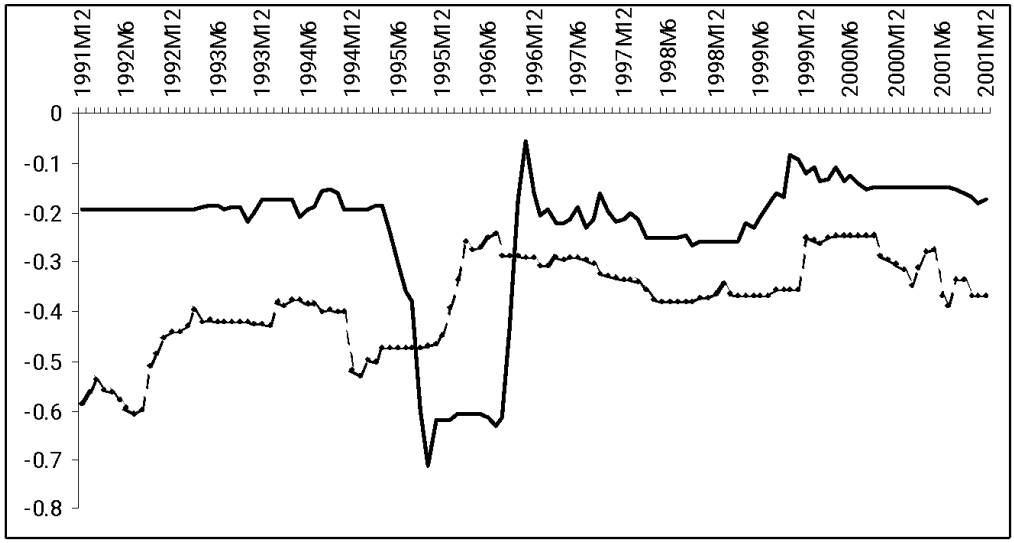

Figure 15 Recursive estimates of speeds of adjustment

I repeated all the calculations using price of petrol expressed in pounds (see Chapter 3.2.2 for details) - results were unchanged. It is clear that increased competition caused by entry of the new station owned by supermarkets and the Pricewatch campaign aimed at competing for the market share with them increased the speed in which the agents responded to shocks caused by falling prices of crude oil and favourable exchange rate movements. It is also visible that change was only temporary.

[7] Due to computation burden more sophisticated methods (i.e. consistent versions of TAR and M-TAR models) could not be used

It seems that introducing the dummies was justified. The Pricewatch campaign and associated price war can be seen as one-off event that affected the values of some estimates but nonetheless did not change the nature of asymmetry in the adjustment towards long run equilibrium.

After ensuring that dummies can and should be introduced I repeated the estimation of the long-run equation and obtained following results:

$$\ln(p)_t = \alpha_1 + \alpha_2 * trend + \beta_1 * \ln(c)_t + \beta_2 * \ln(ex)_t + \sum_{i=1}^{5} \vartheta_i * D_i + \varepsilon_t$$
$$\ln(p)_t = 1.4008 + 0.0004374 * trend + .48936 * \ln(c)_t + .40411 * \ln(ex)_t \qquad (25)$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedasticity	JB Normality
(25)	Value	.76	.54	235.5	219.84	135.38	.0027	4.1995	1.980	1.8984
	P-value					**.000**	.958	.000	.159	.387

Introducing the dummies into (16) did not change the values of the estimates but resulted in the normality of residuals as indicated by JB test statistics. All tests for cointegration were repeated and results were unchanged. Excluding only 5 observations enabled me to test for asymmetric patterns in the transmission with greater confidence.

To test properly for the asymmetry in the long-run speeds of adjustment I used techniques proposed by Granger and Lee (1989). Residuals from (25) were used to test for asymmetry in the adjustment to the long-run equilibrium just as proposed by equation (9). The results are given below

$$\Delta \ln(p)_t = \alpha + \beta_1 * \Delta \ln(c)_t + \beta_3 * \Delta \ln(ex)_t + \pi^+ * \hat{u}_{t-1}^+ + \pi^- * \hat{u}_{t-1}^- + \varepsilon_t$$
$$\Delta \ln(p)_t = -.0064509 + .27928 * \Delta \ln(c)_t + .21439 * \Delta \ln(ex)_{t-1}$$
$$[1.1578] \qquad [8.3636] \qquad [3.9951] \qquad (26)$$
$$-.12740 * \hat{u}_{t-1}^+ - .-.32393 * \hat{u}_{t-1}^- + \varepsilon_t$$
$$[1.6968] \qquad [4.3765]$$

Lagrange Multiplier χ^2 tests

		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedasticity	JB Normality
(26)	Value	.297	1.85	361.61	352.92	17.349	1.937	5.972	1.010	100.424
	P-value					.137	.164	.917	.315	.000

Testing for the symmetry in the adjustment speed was based on simple F test of the null hypothesis of equality of $\pi^+$and π^-. Its value (2.406) allows rejecting null of equal speed of adjustment only at 15% significance level. What is more results of (26) are not consistent with the requirements set by Cook et al. (1998) since the coefficient on positive residuals is significantly different from zero only at 10% not at standard 5% significance level.

In order to get more information about the dynamics of the system the restricted ECM was enriched by lagged explanatory variables. For determining the appropriate lag length the usual Hendry's approach was used.

The most generous model estimated included lags up to order of 6 in case of exchange rate and crude oil prices and 5 lagged differences in retail prices. Therefore to get the overall significance level equal to 5% at each stage of testing-down procedure the significance level for each lagged variable was set at 0.003.

$$1-\prod_{i=1}^{17}(1-0.003)=0.04979 \qquad (27)$$

After the 'testing down' procedure the following model was obtained.

$$\Delta\ln(p)_t = \alpha+\beta_1*\Delta\ln(c)_t+\beta_2*\Delta\ln(c)_{t-1}+\beta_3*\Delta\ln(ex)_{t-1}+\pi^+*\hat{u}^+_{t-1}+\pi^-*\hat{u}^-_{t-1}+\varepsilon_t$$

$$\Delta\ln(p)_t = -.0050829+.27892*\Delta\ln(c)_t+.12725*\Delta\ln(c)_{t-1}+.40727*\Delta\ln(ex)_{t-1}$$

[.95035] [8.6050] [3.9465] [3.1543]

$$-.11119*\hat{u}^+_{t-1}-.-.27891*\hat{u}^-_{t-1}+\varepsilon_t \qquad (28)$$

[1.5443] [3.8907]

Lagrange Multiplier χ^2 tests

		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedasticity	JB Normality
(28)	Value	.358	1.92	369.98	359.56	13.84	1.559	1.927	83025	160.2
	P-value					.311	.212	1.000	.362	.000

Results from (28) also suggest that the speed of adjustment differs according to the nature of the shock to the system i.e. whether upstream prices increased or decreased.

However since the test statistics allow for rejecting the null of symmetric adjustment to the long-run equilibrium only at 10% no ultimate conclusions should be based on this stage of modelling. Results clearly show that there is a need to use more sophisticated techniques to assess the issue of asymmetry.

3.2.2 TAR and M-TAR models

To utilize the TAR and M-TAR models as described in Chapter 1 I had to overcome one problem caused by insufficient developments in the modelling theory.

Techniques described by Enders and Siklos (1999) were not developed to deal with three and more variables, therefore impact of exchange rate could not be evaluated separately.

It was possible to overcome this problem by analysing the price transmission between prices of crude oil expressed in British pounds and net retail prices of 4 Star petrol. Since the price of Brent crude expressed in pounds equals product of exchange rate and price expressed in US dollars, the logarithmic transformation of crude prices in pounds is simply sum of logs of price in dollars and log of exchange rate.

Using prices in domestic currency will be allowed only if in the long run equation coefficients on those two variables are equal to each other. I used the Wald test for testing the null of equality of β_3 and β_4. The value of the test statistic was 2.3304 with associated p-value equal to 0.127. It is lower than the critical values and it does not allow for rejecting the null of equal impact at conventional significance levels. The only problem comes from the presence of the autocorrelation.

Inefficiency caused by the autocorrelation was overcome by the Wald tests based on the Newey-West adjusted variance-covariance matrix with different weights. Using the version with the equal weights and truncation point set at 25% of the sample the test statistic equals 0.74757 which is less than critical value for χ^2 with one d.o.f. (3.84). The Wald test with similar truncation point but different weights i.e. Barlett, Tukey, Parzen gave similar results i.e. 0.58105, 0.55708, and 0.49758, respectively. Changing the truncation point did not affect the result.

I concluded that it is possible to assume that in the level equation the values of the coefficient on crude oil prices and exchange rate is the same so they can be replaced by the new variable, sum of those two.

It is also consistent with results from previous stages of modelling (results from modelling (22))

The same array of tests both diagnostic and for cointegration was applied to the new model. The results are given below.

$$\ln(p)_t = \alpha_0 + \alpha_1 * trend + \beta_2 * (\ln(c)_t + \ln(ex)_t) + \varepsilon_t$$
$$\ln(p)_t = 1.4864 + .003600 * \text{trend} + 47251 * (\ln(c)_t + \ln(ex)_t) \quad (29)$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedascity	JB Normality
(29)	Value	.706	.38	217.15	211.94	159.47	.1137	16.961	2.62	.70
	P-Value					.000	.736	.000	.347	.004

The properties of the new series were thoroughly examined. The ADF test with lag length indicated by information criteria gave the test statistic equal to -2.5364 for levels and -14.6288 for the first differences. Therefore, it is possible to conclude that new series is also I(1).

Typical array of tests for cointegration was also used. To economise on space I will only report the results. The test statistic of the normal test for the residuals DF (such test was recommended by AIC SBC and HQC) was equal to -5.0425. The appropriate critical value at 5% significance level equals -3.8227. Rejecting the null that the residuals are I(1) can be seen as a sign that the series of interest are cointegrated. EG technique in the form given by equation (10) gave the t-statistic on lagged residuals equal to –5.0956 which is greater than the critical values for 1 explanatory variable constant and deterministic tabulated by McKinnon (-4.3919 and -3.820 at 1% and 5% significance levels).

The restricted ECM has an absolute t-value on lagged residuals equal to -4.9583, which exceeds the critical value for a given number of regressors, deterministic trend and a constant, also indicating cointegration.

$$\Delta \ln(p)_t = \alpha + \beta_1 * \Delta(\ln(c)_t + \ln(ex)_t) + \pi * \hat{u}_{t-1} + \varepsilon_t$$
$$\Delta \ln(p)_t = .00250 + .27041 * \Delta(\ln(c)_t + ln(ex)_t) - .17818 * \hat{u}_{t-1} + \varepsilon_t \quad (30)$$
$$[.072292] \qquad [8.2006] \qquad [4.9583]$$

					Lagrange Multiplier χ^2 tests			
R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedascity	JB Normality

(30)	Value	.267	1.89	358.50	353.89	16.926	.08426	7.1043	.57629	86.956
	P-value					.152	.772	.851	.860	.000

Also the unrestricted ECM was used. Results are also given below.

$$\Delta \ln(p)_t = \alpha + \beta_1 * \Delta(\ln(c)_t + \ln(ex)_t) + \pi_1 \ln(p)_{t-1} + \pi_2 * (\ln(c)_{t-1} + \ln(ex)_{t-1}) + \varepsilon_t$$

$$\Delta \ln(p)_t = .25280 + .27123 * \Delta(\ln(c)_t + \ln(ex)_t) - .16708 * \ln(p)_{t-1} + .080131 * (\ln(c)_{t-1} + \ln(ex)_{t-1}) \quad (31)$$

[4.0928] [8.0679] [N/A] [N/A]

		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedasticity	JB Normality
							Lagrange Multiplier χ^2 tests			
(31)	Value	.262	1.90	356.75	349.81	17.02	.297	7.6462	1.6854	89.4852
	P-value					.149	.585	.812	.194	.000

Again this model seems to fit data better than the restricted ECM. The non-standard F-test statistic for join insignificance of lagged level variables equals 11.4338 and exceeds the critical value at 5% significance level i.e. 7.423 (Peseran, Shin and Smith (1996)). The main conclusion is that variables in question form the long run relationship and estimates given above are superconsistent. This not only validates the OLS estimation but also gives the sound basis for testing the asymmetric price transmission.

Keeping all the doubts in mind the simplest TAR model in a form given below was estimated over the sample period. Just like in Chapter 3.2.1 the residuals were taken from the estimation of level equation with 5 dummies to control for the impact of the Pricewatch programme in the first half of 1996, but this time prices of crude oil were expressed in pounds.

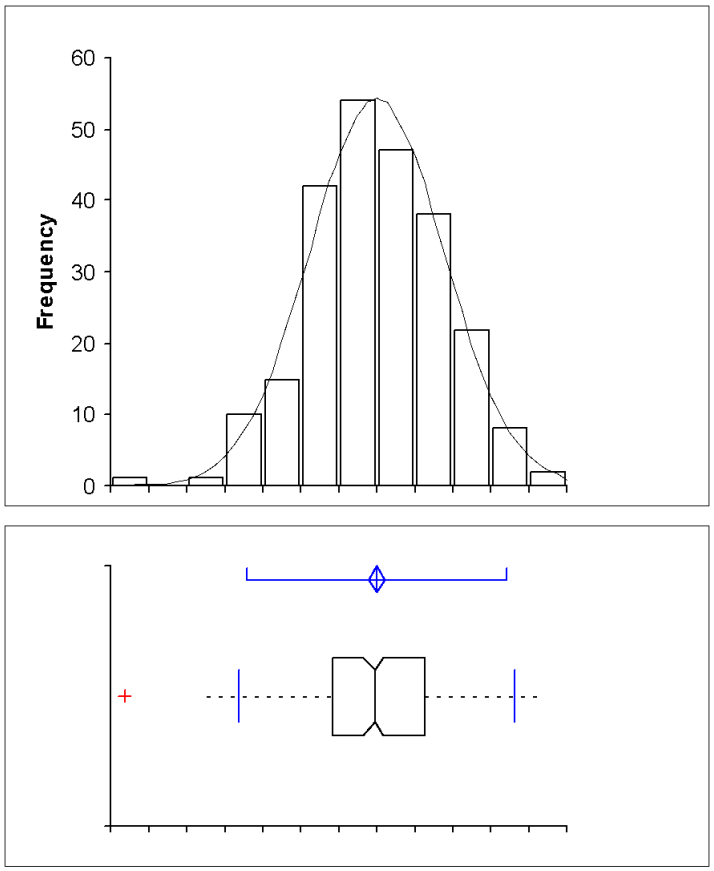

Figure 16 Histogram and box plot of the residuals.

The residuals were divided according to the indicator i.e.

$$I(u_{t-1}) = \begin{cases} 1 \Leftrightarrow u_{t-1} > 0 \\ 0 \Leftrightarrow u_{t-1} < 0 \end{cases}$$

$$\begin{aligned} \Delta\hat{u}_t &= \beta^+ * I(u_{t-1}) * \hat{u}_{t-1}^+ + \beta^- * (1 - I(u_{t-1})) * \hat{u}_{t-1}^- + \varepsilon_t \\ \Delta\hat{u}_t &= \beta^+ * \hat{u}_{t-1}^+ + \beta^- * \hat{u}_{t-1}^- + \varepsilon \\ \Delta\hat{u}_t &= -.21648 * \hat{u}_{t-1}^+ - .32466 * \hat{u}_{t-1}^- \\ &\quad [3.4599] \qquad [5.1982] \end{aligned} \tag{32}$$

							Lagrange Multiplier χ^2 tests			
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedascity	JB Normality
(32)	Value	.141	2.12	332.18	328.71	17.33	1.2948	6.3306	.42671	235.7718

P-value	.138	.255	.899	.514	.000

Threshold set at 0 reflects an implicit assumption that the system under scrutiny behaves differently when upstream prices increase and decrease. The issue of the magnitude of the shocks and its impact on the speed of adjustment will be discussed later.

According to the critical values tabulated by Enders and Granger (1998) the value of the maximum t-statistic (i.e. -3.4599) allows us for rejecting the null of no threshold cointegration between variables at 5% and 1% significance level (appropriate critical values are for 250 observations –2.12 and –2.53 respectively. F-test for the join insignificance of both lagged variables (ϕ test) equals 38.99 and clearly exceeds critical values -5.87 and -8.04, at 5% and 1% significance level.

The hypothesis of equal speed of adjustment could not be rejected by the standard F-test. The test statistic equal to 1.4976 is lower than critical values at typical significance levels.

A clear drawback of modelling asymmetry by using specification described by (32) is that it imposes that characteristics of the adjustment differ only depending on the signs of the disequilibria. The more realistic assumption should be that speed of adjustment might differ if the disequilibria are significant. There are many reasons to predict that this may be the case.

For example, if upstream prices increase by a lot so that margins at all stages of modelling approach zero, agents are more likely to adjust prices toward new long-run equilibrium very fast. But when upstream prices decrease significantly agents may postpone lowering prices and enjoy bigger margins for longer.

I allowed for more complicated process of attraction by considering consistent TAR model as described by equation (13). In such model – consistent TAR the indicator is defined as:

$$I(u_{t-1}) = \begin{cases} 1 \Leftrightarrow u_{t-1} > \tau \\ 0 \Leftrightarrow u_{t-1} < \tau \end{cases}$$

Where τ is the parameter to be estimated. The method employed to find it is based on the one used by modification of method used by Harris and Silverstone (1999). Residuals were divided according to whether they are below or above the potential threshold value. As the potential threshold values I used the percentiles of the residuals. I estimated the 100 models in the form given below.

$$\Delta\hat{u}_t = \beta_1 * I(\hat{u}_{t-1}) * \hat{u}_{t-1} + \beta_2 * (1 - I(\hat{u}_{t-1})) * \beta * \hat{u}_{t-1} + \varepsilon_t$$

The model that yielded the lowest residual sum of square was chosen in order to obtain the preferred value of τ. For all computations I used MS Excel 97.

The analysis suggested that the value of the threshold should be set at –0.05969, in such case RSS equals 0.83589 which is lower than that for model described by(32) (.85388). Figure 17 shows the distribution of the residuals.

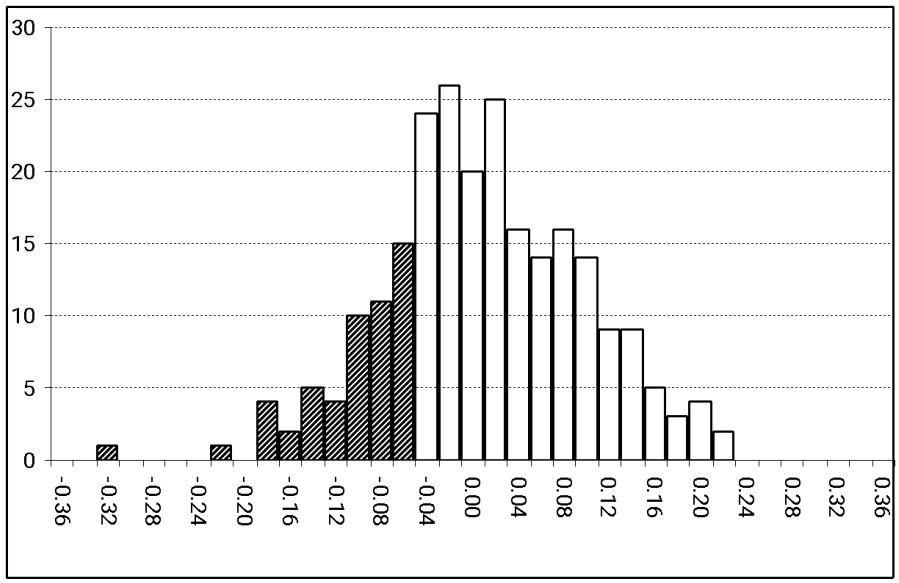

Figure 17 Histogram of the residuals.

Shaded part corresponds to residuals that are below the threshold.

The model with such threshold is described below:

$$\Delta\hat{u}_t = \beta_1 * I(u_{t-1}) * \hat{u}_{t-1} + \beta_2 * (1 - I(u_{t-1})) * \hat{u}_{t-1} + \varepsilon_t$$

$$\Delta\hat{u}_t = -.17884 * I(u_{t-1}) * \hat{u}_{t-1} - .38302 * (1 - I(u_{t-1})) * \hat{u}_{t-1} \qquad (33)$$

[3.0256] [5.8579]

		R^2	DW	AIC	SBC	Auto-correlation	Lagrange Multiplier χ^2 tests: RESET	ARCH	Hetero-scedascity	JB Normality
(33)	Value	.155	2.10	334.111	330.6346	17.5943	.11303	5.971	.32377	244.6854
	P-value					.129	.737	.918	.599	.000

Consistent version is clearly superior to the 'ordinary' TAR. It is indicated by the values of SBC and AIC and by the test statistics.

The t-max* statistic in (33) equals –3.0256 and exceeds the critical values at 5% and 1% (-1.84 and –2.31, respectively) again indicating cointegration.

The F-test for the equality of coefficients has a value equal to 5.366 with p-value equal to 0.021 allowing us to rejecting the null of equal speed of adjustment to positive and negative shocks.

The conclusion from all TAR models above is that the system exhibits slower adjustment to the positive disequilibria that exceed the threshold values. In other words system seems to adjust to the equilibrium in the same manned when small increases and decreases in upstream prices occur but if increases exceed the critical value the adjustment to the equilibrium is much faster.

It is entirely consistent with the results of survey among retailers (mainly independent retailers) in OFT Report (1998: 51). Majority of them claimed that they are reluctant to match their competitors if they operate at very low margin and/or that they feel that their margins are too small. It may suggest that they will use the opportunity to raise margins, by postponing lowering the prices when upstream pries fall but will increase prices very rapidly whenever upstream prices increase. That is why in the latter case new long-run equilibrium is achieved much faster. Their responses clearly indicate that the size of the margin matters – if it is too low they are reluctant to lower the prices to the new long-run equilibrium level. Figure 18 shows simulated adjustment over 24 months after positive and negative shocks which caused 15% discrepancy between current prices and their long-run equilibrium levels.

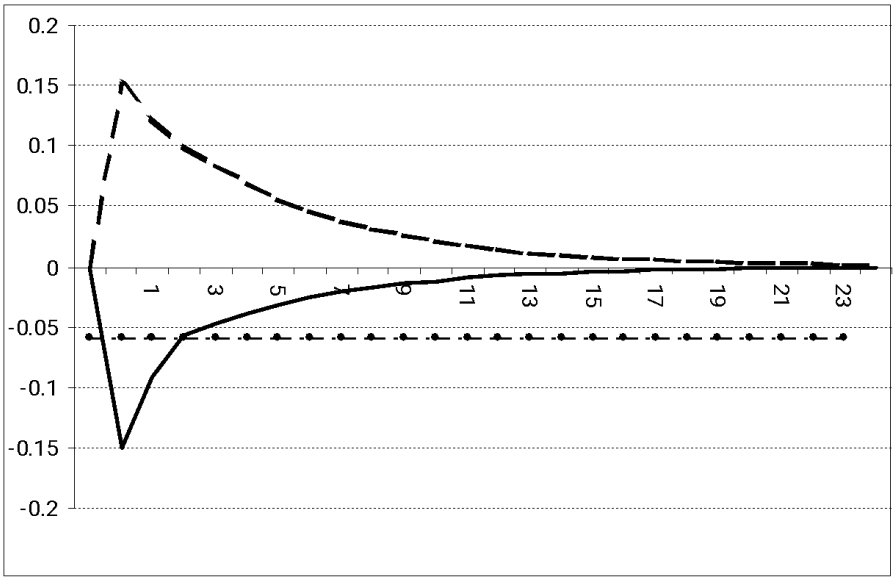

Figure 18 Adjustment to the long-run equilibrium

Dashed line corresponds to the positive disequilibrium caused for example by decrease in the crude oil prices.

This finding can be a basis for the next model in which behaviour of the system depends not only on the direction of the disequilibria but also changes in them affect the adjustment to the long-run equilibrium. In this approach the residuals are not any longer divided according to whether they are positive or negative but whether *change* in shock to the system was negative or positive. In the simplest case the indicator now becomes:

$$I(\Delta u_{t-1}) = \begin{cases} 1 \Leftrightarrow \Delta u_{t-1} > 0 \\ 0 \Leftrightarrow \Delta u_{t-1} < 0 \end{cases}$$

Estimation of M-TAR model gave following results.

$$\Delta \hat{u}_t = \beta_1 * I(\Delta \hat{u}_{t-1}) * \hat{u}_{t-1} + \beta_2 * (1 - I(\Delta \hat{u}_{t-1})) * \hat{u}_{t-1} + \varepsilon_t$$

$$\Delta \hat{u}_t = \underset{[2.4128]}{-.15352} * I(\Delta \hat{u}_{t-1}) * \hat{u}_{t-1} - \underset{[6.2991]}{.38019} * (1 - I(\Delta \hat{u}_{t-1}) * \hat{u}_{t-1} \qquad (34)$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto-correlation	RESET	ARCH	Hetero-scedasticity	JB Normality
(34)	Value	.167	2.12	333.33	329.86	17.895	.38428	6.7575	.08316	247.6153
	P-value					.119	.535	.873	.773	.000

The t-max(M) statistic exceeds the critical values at 5% -1.99 and is very close to the critical value at 1% (i.e. –2.45). The conventional F-test rejects the null of equal speed of adjustment is rejected at 5% and 1% with the statistic equal to 6.68.

As above shows the system adjusts to approximately 15% of the disequilibrium if it follows sudden decrease in the prices of crude oil but 38% of disequilibrium caused by increase in the prices of crude oil of similar magnitude is eliminated over one year.

In light of properties of the M-TAR model described in chapter 1.2, it indicates that sudden decrease in the margin prompts sharp answer but when margins are unchanged or increased adjustment to the new long run equilibrium takes a little bit longer. One can conclude that in the transmission chain agents delay lowering prices when their margins are stable or increasing but changes in the upstream prices that reduce the margin are passed fully onto the consumer.

Again my findings seem to be consistent with the results of the OFT survey (1998:52) Almost half of the retailers responded that they would not match the competition in lowering prices if it means reducing their margin. Their answers also suggest the *change* in the margin is as important as its size.

Also the consistent version of the M-TAR model was estimated. The aim of this model was to account for the possibility that behaviour of the system may depend not only on the sign of the change in the disequilibrium but also on its magnitude. Therefore I estimated the consisted version of the M-TAR model using the first differences of the residuals.

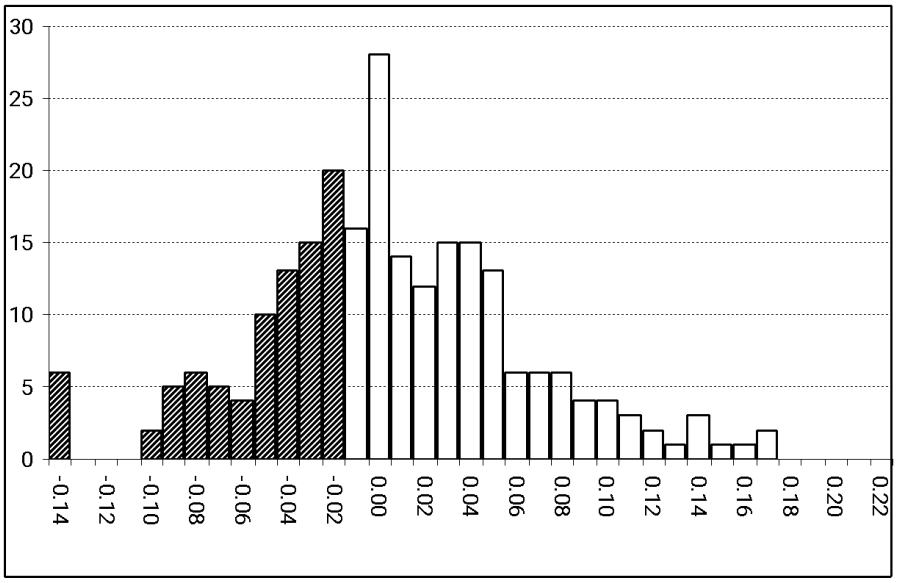

Figure 19 Histogram of the first differences of the residuals.

Shaded part corresponds to the first differences below the threshold.

Indicator was set according to (15):

$$I(u_{t-1}) = \begin{cases} 1 \Leftrightarrow \Delta u_{t-1} > \tau \\ 0 \Leftrightarrow \Delta u_{t-1} < \tau \end{cases}$$

Where τ stands for the threshold. The method of finding it is analogous to that used before in case of consistent TAR model. The estimated model that gave the lowest residual sum of squares (0.82965 compare to 0.83233 in case of ordinary M-TAR) had a threshold equal to -.0034.

$$\Delta\hat{u}_t = I(\Delta\hat{u}_{t-1}) * \beta_1 * \hat{u}_{t-1} + (1 - I(\Delta\hat{u}_{t-1})) * \beta_2 * \hat{u}_{t-1} + \varepsilon_t$$

$$\Delta\hat{u}_t = \underset{[2.3987]}{-.15026} * I(\Delta\hat{u}_{t-1}) * \hat{u}_{t-1} - \underset{[6.3762]}{.38923} * (1 - I(\Delta\hat{u}_{t-1}) * \hat{u}_{t-1} \qquad (35)$$

						Lagrange Multiplier χ^2 tests				
		R^2	DW	AIC	SBC	Auto - correlation	RESET	ARCH	Hetero -sceda scity	JB Normality
(35)	Value	.164	2.12	333.71	330.24	17.95	.35919	6.6967	.05980	251.0294
	P-value					.117	.549	.877	.807	.000

The results of this model are consistent with those from normal M-TAR model. However, as indicated by AIC and SBC it can be seen as superior. The asymmetric response is more visible and the Wald statistic allows for rejecting the null of equal coefficient at 1% significance level (7.4643 with p-value equal to 0.006). The fact that in this case threshold is different from zero and its negative may suggest that change in the margins must exceed specified value in order to prompt fast responses of the agents. If disequilibria changed by less than –.0034, adjustment is weak, but if the change exceeded this threshold (i.e. there was a sudden drop in the value of the

disequilibrium – corresponding for example to sudden increase in the upstream price) response of agents is much quicker and the disequilibrium is eliminated much faster.

3.3 Conclusions

I established by using cointegration approach in conventional and 'threshold' versions that the long-run relationship between crude oil and petrol does exist. Therefore, one can expect that changes in the input prices should be passed to output prices in full in a simple way. However, the results suggest that this process is quite sophisticated.

Speeds of adjustment are clearly different. Techniques used suggest that process of achieving the long-run equilibrium is very complicated and depends on many factors.

The size of the margin is one of them. Agents seem to adjust prices to the new equilibrium level very fast when their margins are low after increases in the upstream prices. In contrast they adjust them slowly when upstream prices decrease, perhaps to enhance their margins. Consistent TAR model suggests that if the discrepancy exceeds critical threshold (around 6%) the adjustment to the equilibrium is rapid, in other cases system adheres to the new equilibrium price slowly.

But not only size of the margin matters, changes in the margin are also important. When they are squeezed agents respond very fast. It is in contrast to times of increasing margins when adjustment is much slower.

Short-run adjustment is also asymmetric. Immediate response to the changes in crude oil prices is bigger for increases than to decreases. What is more, movements in the exchange rate that lead to cheaper crude oil in the UK, are not translated into cheaper petrol immediately.

Recursive estimates suggest that the degree of asymmetry changes with time, but it is hard to say why. In the next chapter I attempt to answer this question.

4 Explanations for Asymmetric Short-run Responses

In the bulk of the literature the stress was put on detecting asymmetric patterns not on explaining it. The first attempt to move a little further and distinguish between sources of those asymmetric patterns was by Berenstein et al. (1997). They identified three hypotheses that may explain departures from symmetric adjustment of retail prices to changes in crude oil prices – oligopolistic co-ordination equilibrium, inventory costs and short-run market power caused by the search costs that consumers face.

Those explanations differ in their implications for margins at different points in the distribution chain (e.g. short-run market power occurs at retailers' level, inventory costs at refineries' level, etc.). Therefore authors were able to differentiate among them on the basis of econometric analysis of transmission. If asymmetry was present at one stage but absent at the other authors concluded that explanations which focus on that stage are probably true.

This approach has many virtues. Firstly, it is simple, but the results are suggestive and credible. Secondly, if some additional information on e.g. competitive structure along various points in the distribution chain is available, the analysis can be used for policy purposes. Thirdly, if the data are extremely detailed, and a market under scrutiny is distinct, researcher may test very sophisticated theories.

The best example is Eckert (2002). Using weekly petrol prices in Windsor, Ontario, Canada he proved that although retail prices do respond faster to wholesale price increases than to decreases, they also exhibit a cyclic pattern inconsistent with any common explanations of response asymmetry. He explained them by the alternating-move price setting oligopoly model (see Eckert (1999) for details). The really interesting point is that what seems to be asymmetry can really be a sign of battle over the market share. Therefore the competition policy may be welfare decreasing. The only problem with this method is that it requires very detailed data about one distinct market.

The second way of testing the explanations for asymmetry was proposed by Peltzman (2000). He examined 242 commodities in the period from 1978 until 1996 and tried to find the link between degree of asymmetry and characteristics of the market. He found a negative correlation between strength of asymmetry and input price volatility and no correlation with proxies for inventory costs, asymmetric menu costs of price changes and imperfect competition.

Until now it is the only study aimed at explaining asymmetry by cross-market comparisons. Also in this case the availability of the data is the main problem.

Of course, one cannot apply his explanations to the market for petrol in the UK directly. However it can be of some use. For example, I can try to establish whether one of his general conclusions about the nature of the asymmetry applies to the UK.

I will start the analysis of possible sources of asymmetry with the most popular explanations connected with imperfect competition and market power. Then I will proceed with more benign explanations based on the inventory costs and production lags. I will have a closer look at one of Peltzman's (2000) predictions about the general characteristics of asymmetry in the short run adjustment.

I do not aim at testing any particular theory; instead I try to show only some patterns in the data that *may* be consistent with some explanations. Since so far explanations were dealing with the short run adjustment as described by (8) I try to apply them to the short-run responses described in 3.1.

4.1 Explanations connected with the issue of market power

Most of the literature on the asymmetric price transmission was inspired by inquires made by authorities into the pricing policies of petrol retailers and wholesalers (see Chapter 1.1 for details).

It may be caused by the fact that asymmetric response is not consistent with the idea of perfect competition. In the best of the worlds competition forces all agents to lower the prices as soon as possible. The response to decreases and increases in the upstream prices should be symmetric and should perfectly reflect changes in costs. Most economists dream about such world and any sign that it may be the case disturbs them.

Not only them. For most laymen, lack of perfect competition means something bad and it takes little time before word 'collusion' is said.

Unfortunately no theory has been developed to deal directly with those issues. Balke et al. (1998: 4) stated that

> "....no formal model shows a relationship between the market structure and asymmetric response of downstream prices to changes in upstream prices. Informal models usually involve firms that are concerned about maintaining a tacit collusion".

Recently some sophisticated theories that do not attribute the market power to the structure of the market directly but explain it as a result of the search costs that consumers face were developed.

Let's start with the informal models and then move to more formal explanations. Consider an industry with a few dominant firms that are engaged in collusion. One way to overcome problems with maintaining the collusive agreement is connected with reputation. When firms have imperfect knowledge of the upstream prices other firms pay and they want to adhere to the agreement then they face an asymmetric loss function. When the upstream prices increase each firm increases its selling price to show others that it adheres to the agreement. When the upstream prices fall, firms may postpone lowering their prices so as not to send a signal to its fellow-conspirators that it cuts the margins and reneges from the agreement. This explanation is very difficult to test. For a discussion about those issues see Kühn (2000).

Another explanation involves short-run market power for retailers. Each outlet has a local market power that is limited only by consumer search. The amount of search depends on benefit-cost analysis. When consumer knows that retail prices are currently volatile she may be

more likely to believe that an increase in one station's price reflects crude oil price changes, rather than a change in the station's relative price. Thus, the expected gain from search may be smaller. Each retailer knows that it implies a temporary decline in the elasticity of demand and increases its margin. This may dampen the rate of passthrough of upstream prices decreases and exacerbate the passthrough of increases.

Bénabou and Gertner (1993) formalised a theory of costly endogenous search and concluded that economy-wide inflation or common shocks to prices can increase or decrease the equilibrium amount of search. This may increase or decrease competition among sellers. They found that search is more likely to decrease if the cost of search is high to begin with. This assumption seems to be appropriate for petroleum products, although author knows no formal study about the cost of search for cheaper petrol.

The theory of costly search applies to retailers' margins and it has not much to say about refiners' or wholesalers' margins. Therefore this theory is aimed at looking for asymmetry in the short-run response. If the theory is true then asymmetry should be greater in months of volatile petrol prices or high inflation.

To see whether this explanation applies to UK market I decided to amend the model given by (8) by introducing some slope dummies. Introducing them allowed me to distinguish between asymmetry in months when retail prices were volatile from asymmetry in months when search should be more profitable. Such amended model is given below:

$$\begin{aligned}\Delta \ln(p)_t &= \alpha_1 + \alpha_2 * trend \\ &+ D * \beta_{1,0}{}^{+} * \Delta^{+} \ln(c)_t + \hat{\beta}_{1,0}{}^{+} * \Delta^{+} \ln(c)_t + \beta_{1,0} * \Delta \ln(c)_t + \beta_{1,1} * \Delta \ln(c)_{t-1} \\ &+ D * \beta_{2,0}{}^{+} * \Delta^{+} \ln(ex)_t + \hat{\beta}_{2,0}{}^{+} * \Delta^{+} \ln(ex)_t + \beta_{2,1} * \Delta \ln(ex)_{t-1} \\ &+ \pi_1 * \ln(p)_{t-1} + \pi_2 * (\ln(ex)_{t-1} + \ln(c)_{t-1}) + \varepsilon_t\end{aligned} \tag{36}$$

The only problem was connected with defining periods of high volatility. I decided to compute monthly changes in petrol prices and compare them to specified threshold. When monthly change exceeded this threshold I assumed that the volatility was high. As a threshold I used the standard deviation of monthly changes. Using mean of absolute changes gave broadly same results but fewer observations were below the thresholds.

The dummy variable D in (36) was defined as:

$$D = \begin{cases} 1 \Leftrightarrow |\eta| < \theta \\ 0 \Leftrightarrow |\eta| > \theta \end{cases} \tag{37}$$

Where η stands for the monthly rate of change in the prices of petrol and θ represents threshold (100%, 50%, 33% and 25% of the standard deviation of monthly changes).

The product of dummy defined in this way and positive changes in explanatory variables together with its standard error gave basis for testing whether in periods of low volatility of petrol prices asymmetry is smaller. Results of the estimations are given below. [8]

Threshold chosen	Variable	Coefficient	Threshold chosen	Variable	Coefficient
100%*sd	D*Δ^+ln(c)	**-.31344*** **	100%*sd	D*Δ^+ln(ex)	-.45059
	Δ^+ln(c)	**.31804*** **		Δ^+ln(ex)	**.74916** **
50%*sd	D*Δ^+ln(c)	**-.30985** **	50%*sd	D*Δ^+ln(ex)	-.48676
	Δ^+ln(c)	**.24947*** **		Δ^+ln(ex)	**.66824** **
33%*sd	D*Δ^+ln(c)	**-.26191** **	33%*sd	D*Δ^+ln(ex)	-.53253
	Δ^+ln(c)	**.22689*** **		Δ^+ln(ex)	**.69439** **
25%*sd	D*Δ^+ln(c)	**-.29521** **	25%*sd	D*Δ^+ln(ex)	-.49066
	Δ^+ln(c)	**.22371*** **		Δ^+ln(ex)	.61399

Table 5 Asymmetry and volatility of petrol prices

The results in the third column clearly show that the asymmetric short run response is weaker or even non-existing when petrol prices are very stable. This can be caused by the fact that changes in prices in those periods are very small and they can be invisible in my aggregated data but it also can be seen as a support for the hypothesis that short-run market power is responsible for asymmetry.

In my opinion, results based on harsh criteria (low values of threshold) should be interpreted with care. It is possible that not enough observations were used to compute the value of the coefficients so they could not represent true characteristics of the system.

It is also possible that the lack of asymmetry in some months was caused not by active consumer search but by introducing Pricewatch campaign and intense competition at its beginning. Unfortunately, I cannot distinguish between those two with certainty.

[8] Notation used: * significantly different from zero at 10%, ** at 5%, at less than 1%, **** Microfit 4.0 reports p-value equal to .0000

To account for this problem I repeated all the computations for smaller sample (January 1982- December 1994). Results are given below. Again those for harsh criteria may not represent true characteristics of the system as not enough (less than 20-30 observations) were used:

Threshold chosen	Variable	Coefficient	Threshold chosen	Variable	Coefficient
100%*sd	$D*\Delta^{+}\ln(c)$	**-.17827***	100%*sd	$D*\Delta^{+}\ln(ex)$	**-.66689****
	$\Delta^{+}\ln(c)$	**.24613*****		$\Delta^{+}\ln(ex)$	**.77514*****
50%*sd	$D*\Delta^{+}\ln(c)$	**-.27998***	50%*sd	$D*\Delta^{+}\ln(ex)$	**-.65215***
	$\Delta^{+}\ln(c)$	**.20822*****		$\Delta^{+}\ln(ex)$	**.57161****
33%*sd	$D*\Delta^{+}\ln(c)$	**-.29390***	33%*sd	$D*\Delta^{+}\ln(ex)$	-.51737
	$\Delta^{+}\ln(c)$	**.21379*****		$\Delta^{+}\ln(ex)$	**.46911****
25%*sd	$D*\Delta^{+}\ln(c)$	**-.29505***	25%*sd	$D*\Delta^{+}\ln(ex)$	-.46505
	$\Delta^{+}\ln(c)$	**.20435*****		$\Delta^{+}\ln(ex)$	**.47215****

Table 6 Asymmetry and volatility of output price 1982-1994

Again the third column clearly shows that asymmetric responses to changes in the upstream prices caused by crude oil prices are smaller when prices of petrol are stable. For example, when volatility is small (below one standard deviation of monthly changes) over the whole sample 1% increase in the crude oil prices causes almost the same change (in absolute values) in petrol prices as 1% decrease. Wald test cannot reject the null that the sum of the coefficients equals zero (so there is symmetric transmission when prices are stable).

When volatility is high, retailers pass the increases much faster. In months when prices of crude oil increase by 1% retailers change their prices (in absolute values) by more (exactly by 0.24%) than they change it when crude oil prices decrease by 1%.

The results about asymmetry caused by movements in the exchange rate are slightly different. In some cases the coefficients aimed at capturing the effects of small volatility are not statistically different from zero. This can suggest that movements in the exchange rate have small impact on the retail prices. In such case, even if the asymmetry existed I was not able to detect it using my aggregated data.

It is also possible that the fact that asymmetry caused by the exchange rate is present at stages when prices are expressed in dollars (mainly at the refinery stage). In such case, if the transmission of prices from refinery to retailers takes more than one month, it will not be visible in the data using the model defined as (36).

To account for economy-wide inflation I repeated the computations but using the rates of monthly changes in the RPI as a proxy for volatility. The signs of the coefficients on the variables aimed to capture the effect of volatility were of expected sign but they were not significantly different from zero at 5%. It may be the sign that drivers pay attention to volatility of prices of petrol regardless of economy-wide inflation.

Another explanation that is connected with the issue of market power is a variant of the trigger price model of oligopolistic coordination (see Green and Porter (1984) for details). In that model each firm restricts its output to a level below the competitive (Nash) output if, and only, if the market price is above particular threshold.

Imagine a firm (refinery, wholesaler or petrol station), which chooses its selling price with imperfect information about prices charged by its competitors. A firm charges price above competitive (Nash) level if its sales remain above the threshold level. When the sales drop and margins are squeezed firm sees it as a sign that competitors cut the price so the optimal response is to do the same.

The asymmetry can be caused without any collusive behaviour. It can arise because positive upstream price shock (effect of the unfavourable exchange rate or increases in the crude oil prices) trigger downstream price increases, otherwise, margins will be negative[9].

Downstream price need not to drop immediately in case of negative shock if sales and margins are not changed immensely. Prevailing prices are treated in such case as a focal point for oligopolistic coordination.

Over time prices return to their long-run equilibrium level because random shocks in demand make firms cut their prices in response to the threat of competition.

This theory cannot be used to attribute the asymmetry to a particular stage in the transmission chain, because at all of them the mechanism works in the same way. Distinction is possible only when detailed data on prices prevailing at different stages and on characteristics of those stages are available.

But it may be reasonable to assume that this theory should explain asymmetry at retailers' level at which sellers are not price takers, buyers (drivers) are not completely informed about the price of each seller and information on other retailers' sales and prices is not easily available.

[9] This part of the theory is completely consistent with the findings of the Office of Fair Trading (1998) about the thin margins in the UK market.

Therefore patterns consistent with this explanation should be visible in the short-run adjustment as described in (8).

This explanation is very sophisticated and testing it requires tremendous amount of data on the market structure, margins at different stages in transmission, etc.. Since I was not able to collect such detailed data I was not able to test the theory directly. However, using my data I could check whether changes in the margin really change the degree of asymmetry in the short-run responses.

I decided to check whether *changes* in the proportion of input cost in the retail prices might somehow affect asymmetry. As the proxies for input costs I used crude oil prices expressed in pounds and producer price index.

The idea is that when the proportion of input costs (upstream price) increases the margins are squeezed. That can indicate 'hard times' for producers, wholesalers and retailers. One can think that in those months sales can fall enough so that response can be 'triggered' and asymmetry should be less visible. One additional problem is that agents can accustom to the 'hard times' so if margins were being squeezed for the longer time it may not have the same effect as one-off, sudden change. Below I describe how I accounted for it.

I estimated equation given by (36) but this time the dummy was defined in a different way. This time the value of the dummy depended on the change in the *proportion* of upstream prices in the retail price. To account for trends in the changes of the proportion of input costs I used different orders of the moving averages of the changes in the cost share.

The dummy introduced was defined according to (38)

$$D = \begin{cases} 1 \Leftrightarrow \mu > 0 \\ 0 \Leftrightarrow \mu < 0 \end{cases} \tag{38}$$

Where μ stands for the moving average of the order k of the monthly changes in the share of crude oil prices in the retail price $[(ex_t*c_t)/p_t - (ex_{t-1}*c_{t-1})/p_{t-1}]$. k=1 (just monthly changes), 2, 3, 4, 6, 9, 12.

The results of the estimations are given below. The Wald test statistic is for the null that coefficients are equal but have opposite signs (there is no asymmetry):

	Order of the moving average (months)

	1	2	3	4	6	9	12
D*Δ⁺ln(*c*)	**-1.5080*** **	**-.5893****	**-.27248*** **	**-.22517* **	**-.27649*** **	**-.19268** **	.087011
Δ⁺ln(*c*)	**1.7026*** **	**.69041*** **	**.37449*** **	**.39351*** **	**.30135*** **	**.26011*** **	.10199
Wald test	**8.8321**	2.1326	1.9022	**5.6005**	.080663	.57330	**6.1888**
Σ	**0.1946*** **	0.10111	0.10201	**0.16834** **	0.02486	0.06743	**0.189001* **
D*Δ⁺ln(ex)	**-.82746** **	-.50500	-.59839	**-.64918* **	-.18805	-.48733	-.33160
Δ⁺ln(ex)	**1.1654*** **	**.79303** **	**.85855*** **	**0.93094** **	**.58515* **	**.79974** **	**.63026* **
Wald test	2.3278	1.4929	1.0442	1.2724	2.5947	1.6274	1.3387
Σ	0.33794	0.28803	0.26016	0.28176	0.3971	0.31241	0.29866

Table 7 Asymmetry and change in the cost share

Results of the Wald test are not systematic but for k=1,4,12 one can conclude that asymmetry in short-run response to the crude oil prices is present even when input cost starts to constitute growing share of the retail price. In those months the difference between responses of retail prices to 1% increase and decreases in the crude oil prices equals 0.17-0.19%. In other months degree of asymmetry is much greater.

To make sure that the results were credible I repeated all calculations but I used the index of producer prices as a proxy for all other costs. Of course such proxy is extremely 'crude' but nonetheless one can argue that it can be useful in capturing the effects of costs of labour, machinery, etc.. Data was obtained from the National Statistic web page. Results were unchanged.

	Order of the moving average (months)						
	1	2	3	4	6	9	12
D*Δ⁺ln(*c*)	**-.46154****	**-.39399****	**-.39752****	**-.28915*** **	**-.27581*** **	**-.29693*** **	**-.28486*** **
Δ⁺ln(*c*)	**.31300****	**.29976****	**.28767*** **	**.25517*** **	**.24464*** **	**.24961*** **	**.26309*** **
Wald test	2.3913	.87279	1.1997	.12011	.093986	.21938	.051312
Σ	-0.1485	-0.09424	-0.10983	-0.03398	-0.03117	-0.04732	-0.02177
D*Δ⁺ln(ex)	**-1.151****	**-.79881** **	**-.59560* **	-.064917	-.20622	.025927	-.10043
Δ⁺ln(ex)	**1.0695****	**.85512****	**.66363*** **	**.46356* **	**.46442* **	36230	.43415
Wald test	.090797	.033954	.044605	2.1069	.65950	1.5412	1.6988
Σ	-0.0815	0.05631	0.06803	0.398643	0.2582	0.388227	0.33372

Table 8 Asymmetry and the cost (PPI) share.

This time the Wald test in none of the cases allowed for rejecting the null that the sum of the coefficients is equal to zero. From those results one can conclude that the asymmetry seems to depend heavily on the change in the margin. But again using such crude proxy for producers' costs may affect the results. I can only but hope that other studies involving detailed data will deal with those problems in more professional manner.

Again results about asymmetry caused by the exchange rate movements are not systematic. It is possible to attribute it to the fact that asymmetry caused by the exchange rate occurs mainly at refineries' level and the passthrough takes some time so it may not be visible in month when increase occurred.

Signs of the coefficients and their significance in Table 7 and Table 8 suggest that when the margins are squeezed asymmetric response is smaller. In other words, petrol prices change in the same manner when prices of crude oil change. This can be treated as the evidence that firms postpone lowering prices in favourable market conditions treating existing prices as 'focal point". In months when margins decrease the response seems to be symmetric. It can suggest that in times when margins were squeezed responses were 'triggered' more often and as a result asymmetry in the short-run adjustment was less visible.

4.2 Other explanations

Although the existence of asymmetry could be consistent with market power, there are other explanations. They include inventory management, accounting practices and refinery adjustment costs.

Inventory management can easily be a source of asymmetry. Inventories cannot be negative and at some point the cost of *decreasing* inventories must increase substantially. With increases in inventories the situation may be different.

Such asymmetric inventory adjustment cost caused only by excess storage capacity and non-negativity constraint on inventories may give rise to asymmetric short-run response (see Borenstein and Shephard (1996) for a theoretical model).

For the UK market OFT (1998) and claimed that all the refineries are operating below the capacity. It is obvious that they also have excessive storage capacity. In such case it is possible

that marginal costs of storage adjustment may be decreasing at low and constant at high levels of reserves. Therefore inventory adjustment costs may be responsible for asymmetry. [10]

The asymmetry caused by changes in inventories may be strengthened by FIFO accounting. If inventories fall when upstream supply falls, firms sell the products that incorporate the higher upstream prices sooner. If inventories rise when upstream supply increase (e.g. because refineries buy more cheaper crude), firms sell the products that incorporate lower prices later. Of course, with LIFO accounting the opposite is true and firms should lower product prices more swiftly when upstream prices fall and supply increases.

To assess the value of this explanation one must have data about accounting practices and behaviour of inventories for refineries, wholesalers, etc.. However, it is enough to say that FIFO accountancy is used quite commonly so it can be responsible for asymmetry.

Assessing the impact of inventory costs would require data on storage capacity and its utilization in every refinery. Such information is unavailable on monthly basis. What is more, testing whether it really has some impact on asymmetry would be constrained to distinct markets. Unfortunately, I could not do it with my data.

Asymmetries can also arise at refineries' level in a different way. Those huge enterprises face high costs of changing their output and whenever possible they avoid sharp changes. However, in times of sudden developments in the world markets quick and costly response is inevitable. Costs of such reduction in production can, in principle, lead to asymmetric patterns in the short run response to changes in upstream prices.

But this theory can be used only under some conditions. What is an obvious, cost of adjusting output must be significant. But the OFT (1998) report clearly states that refineries are producing below capacities so that although they have excess storage capacities they still have huge inventories of both finished products and crude oil. This should make them resistant to volatile market – sudden changes of crude oil prices do not mean the necessity to decrease production.

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[10] Extremely straightforward and general discussion on connections between price rigidity and inventory costs, which can be seen as additional support for this hypothesis can be found in Blinder (1982). For our purposes it is enough to say that when crude oil prices increase and firms cut their purchases in response to higher upstream prices, according to Blinder (1982) asymmetry may be result of the behaviour of *any* profit-maximising firm.

What is more, if they work below capacity, costs of reducing output may not be significant – because some machinery is already unused, some workers were fired before, etc.. This neat explanation has its drawbacks and does not seem to be plausible.

Not only stylised facts imply it. Peltzman (2000: 489) pointed out that asymmetry should be greater when degree of input price volatility is small. He found that negative effect of input price volatility on asymmetry is empirically important.

He concluded that this should not be a surprise if one knows basic laws of economics. The law of demand implies that lowering prices increases sales. The 'price' of asymmetric response is equal to the margin firms can make on extra sales. This price is greater when input prices are more volatile.

To account for the possibility that asymmetry can be greater when volatility of input prices is small I estimated (36) over whole sample with dummy variables defined so as to capture the effect of volatility of crude oil prices expressed in dollars and pounds. The dummy variable was defined according to (39):

$$D = \begin{cases} 1 \Leftrightarrow \kappa < \xi \\ 0 \Leftrightarrow \kappa > \xi \end{cases} \tag{39}$$

Where κ stands for the moving average of the order k of the monthly changes in the prices of crude oil, ξ stands for the value of such change assumed to be a threshold [k=4,6,12,24 and ξ=100%, 50%, 33% and 25% of the standard deviation of the difference between monthly changes and the appropriate moving average of the monthly changes].

Table 9 and Table 10 give the results for prices expressed in dollars and ponds.

Variable	Threshold chosen	Order of the moving average (months)			
		4	6	12	24
$D*\Delta^{+}\ln(c)$	100%*sd	-.13868	-.038301	.096858	.0034015
	50%*sd	**.29037***	.19225	**.80527*****	**.48857****
	33%*sd	-.046421	**.59975****	**.73350*****	**.71831***
	25%*sd	-.071082	**.73434*****	**.84391*****	**.73606***
$D*\Delta^{+}\ln(ex)$	100%*sd	**1.1747****	**1.2686***	.54813	**1.6607****
	50%*sd	.58554	.44811	.35483	.38672
	33%*sd	.55631	.27974	.40974	.39402
	25%*sd	.59421	.36507	.31691	.34056

Table 9 Asymmetry and volatility of input prices.

Crude oil prices expressed in USD.

Variable	Threshold chosen	Order of the moving average (months)			
		4	6	12	24
$D*\Delta^{+}\ln(c)$	100%*sd	**.28784****	-.029048	.10929	.085175
	50%*sd	.068445	.14235	**69521******	**50623****
	33%*sd	.031791	.28379	**.80858*****	**.73631*****
	25%*sd	.072157	23850	**.78320*****	.43913
$D*\Delta^{+}\ln(ex)$	100%*sd	.17814	14560	-.054419	.26649
	50%*sd	.15151	.090130	.11625	.11330
	33%*sd	.17111	-.048512	-.0059049	.14640
	25%*sd	.21760	-.15403	.10364	.24903

Table 10 Asymmetry and input price volatility

Crude oil prices expressed in pounds. Author's calculations.

Results are not systematic and no final conclusions should be based on them. However, it seems that asymmetry is lower when input prices are volatile. This is consistent with Peltzman's predictions. Although it is hard to distinguish between sources of asymmetry, my results do not imply that cost of adjusting output in refineries is among most convincing explanations.

Findings of OFT also indicate that in the last 20 years, when refineries in the UK were producing at less-than-full capacity, additional reduction in production because of the higher prices of crude oil was not costly enough to create huge disturbances in the transmission chain.

One final explanation usually quoted in the literature about price rigidity is connected with the menu costs. Lowering prices is always associated with certain costs (re-labelling, etc.). Fast response to input price decreases may be seem as a waste of menu costs when input prices are rising and input price increases seem to be more persistent than input price decreases.

For evaluating importance of this hypothesis I computed the first difference of prices as the proxy for the drift of input prices over the sample period. Then the k-th order moving averages of this drift were calculated (k=4,6,12,24) and compared to zero. When it was positive it was assumed that input prices were drifting upward so the menu-cost story should imply greater degree of asymmetry.

To measure that, the dummy variable was again introduced. It was defined according to (40)

$$D = \begin{cases} 1 \Leftrightarrow \psi > 0 \\ 0 \Leftrightarrow \psi < 0 \end{cases} \qquad (40)$$

Where ψ stands for the value of monthly change in the log of crude oil prices.

It is impossible to say that drift may in any way be a proxy for menu costs. It can be only seen as a way to distinguish months when menu cost *could* be important in explaining asymmetry.

Equation (36) was again estimated and results for log of the prices expressed in dollars and pounds are given in Table 11 and Table 12.

Variable	Order of the moving average (months)			
	4	6	12	18
$D^*\Delta^+\ln(c)$	.16373	.073351	**.19101***	**.20869****
$\Delta^+\ln(c)$	.031563	.10803	.018226	013283
$D^*\Delta^+\ln(ex)$	-.33505	-.076975	-.023972	.16958
$\Delta^+\ln(ex)$	**.52638****	.43546	.40906	.32747

Table 11 Asymmetry and drift of input prices I

Crude oil prices expressed in dollars. Author's calculations.

Variable	Order of the moving average (months)			
	4	6	12	18
$D^*\Delta^+\ln(c)$	**.23226***	.11941	**.22241****	**.19738***
$\Delta^+\ln(c)$	-.029489	.073387	-.0040224	.023078
$D^*\Delta^+\ln(ex)$	-.21490	-.36292	-.27049	-.080715
$\Delta^+\ln(ex)$	.55805	**.65687***	**.52835***	.47293

Table 12 Asymmetry and drift of input prices II

Crude oil prices expressed in pounds. Author's calculations.

It is important to notice that:

- Only in two cases are the coefficients on variables aimed at capturing the effect of the drift significantly different from zero at 5% in Table 1. For crude oil prices expressed in pounds only three coefficients in Table 12 were significantly different from zero;
- Signs of the coefficients are neither stable nor consistent (e.g. Table 11 fourth row). They change according to the order of moving averages. Closer analysis revealed that excluding few initial or final observations changed the results dramatically;
- The values of the coefficients are not consistent- results reported in the fourth column in Table 11 varies from -.33 to +0.17;

Of course I cannot state that menu costs are irrelevant and this explanation does not apply to the UK. I merely say that results obtained in a very crude way using aggregated data are not consistent.

What is more, costs of changing the prices are not very big. On the refinery level prices change every day if not every hour, wholesalers are also accustomed to changes. Re-pricing petroleum products on retailer's level usually involves changing the numbers displayed near the site. It is surely not that expensive.

4.3 Conclusions

Among possible sources of asymmetry those connected with short-run market power seem to be confirmed by the data. Together with the findings from Chapter 3, which clearly indicate that asymmetry depends on the size of the market margin and changes in it, one can conclude that agents' behaviour is responsible for short- and long-run asymmetry. Exactly on which stage it occurs is less clear. Results about the asymmetry caused by the exchange rate movements are not clear, it is possible that the nature of asymmetry at the refinery level is somehow 'special'. Establishing exactly why and how must be done on the basis of very detailed data.

Concluding Remarks

The aim of this piece of work was to discuss (mainly form the empirical point of view) the issue of asymmetry in the UK market for 4 Star petrol during last 20 years.

My results confirm that petrol prices respond in a different manned to increases and decreases in upstream prices, regardless if they are caused by changes in the crude oil prices on international markets or by movements in the exchange rate.

The immediate response to increases in crude oil prices is almost twice as big as response to decreases of the same magnitude. Movements in the dollar-pound exchange rate, which should be beneficial to consumers do not seem to passed to them immediately unlike the movements that increase the price of crude oil expressed in pounds.

I was also one of the first researchers (perhaps event the first) to employ TAR and M-TAR models to British market. Results suggest that adjustment to the long-run equilibrium differs according to the characteristics of the shocks to the system. In particular M-TAR model pointed out that changes in the market margin have great influence on the speed of adjustment. This was confirmed by the results of asymmetric ECM used to analyse the short-run responses.

TAR models both in the normal and consistent versions also indicate that profitability of the economic activity has the impact on speeds of adjustment to the long-run equilibrium. This also can be seen as my contribution to the economic theory so far no study attempted to explain what causes asymmetries in the UK market for petrol.

The importance of the behaviour of the firms at different stages of transmission seems to be a major factor influencing degree of asymmetry. Alternative explanations, suggesting that factors beyond firms' control (inventory costs, menu costs, etc.) are to blame, do not get strong and consistent support. The size of the margin and changes in it seem to affect asymmetry in a more powerful manner.

My results (especially those based on the ECMs) are consistent with the bulk of the literature on the UK market. However, I suggest that the problem is far from being solved and requires more attention, regardless of our views on the asymmetry.

If we argue that asymmetries are welfare-decreasing (in the sense that consumers do not benefit from reductions in the input prices as soon as possible) then analysing the explanations behind this phenomenon is necessary for policy purposes.

My results suggest that affecting the behaviour of the agents is *the* way to do it. It was shown that one short price war and entry of new players in the market reduced asymmetry. This seems to confirm old view that competition is the best way to enhance consumers' welfare.

But even if we believe that consumers should keep on complaining on the asymmetry then more work is necessary. My dissertation confirmed Peltzman's view – economic theory of price transmission is far from perfection. New theories should be created, if not to improve the world, then to explain it.

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